ENTERPRISE NETWORK DESIGN & IMPLEMENTATION FOR IRONONE - KANDY BRANCH

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1. Introduction

1.1. Problem overview

The problem involves designing and implementing a network infrastructure for IronOne's branch in Kandy according to the given department plan

Department	Customer care	Sales and Marketing	Finance	Legal	HR	Developers	Network Team	Server room
No. of Users	10	20	25	5	10	55	5	5

The goal is to separate each department with a unique subnet and ensure that they cannot communicate with each other unless there is a special requirement.

The provided IP address scheme allocates 192.168.10.0/24 for all departments except the server room, which should have the subnet 10.254.1.0/24.

To address this problem, the following tasks need to be completed:

1. Design a VLAN and IP subnetting scheme based on the given requirements and address allocation.
2. Prepare a network design that includes the necessary network devices and connections to implement the VLANs and subnets.
3. Implement the network infrastructure according to the designed plan, including configuring VLANs, IP addressing, routing, and security measures.
4. Conduct verification tests, such as ping, trace route, and telnet, to ensure proper network connectivity and isolation.
5. Provide recommendations for potential future enhancements to the networked system, considering redundancy, security, and scalability.

The overall objective is to create a robust and secure network infrastructure that enables efficient communication within departments while maintaining isolation between them, except when required.

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2. Proposed IP plan allocation

The technique used to prepare the IP allocation plan for this company is the Variable Length Subnet Masking (VLSM). Using VLSM, we can allocate subnets with varying sizes according to the number of users in each department; ensuring that each department receives an appropriate number of IP addresses without wasting address space.

By assigning unique subnets to each department, we can meet the management's requirement of separating departments and preventing communication between them, except when necessary. This improves network security and enhances the overall management of the network.

VLSM also enables better scalability and future growth. By this allocation plan, we can easily accommodate additional users in each department by adjusting the subnet sizes without requiring major reconfiguration or IP address changes throughout the network.

Using VLSM, Here's a proposed and suggested IP allocation plan using VLANs and subnetting. While creating the plan, assumptions are made and defined below;

Note: The provided IP allocation plan assumes that the VLAN IDs and subnetting scheme are within the acceptable range and do not conflict with any existing network configurations.

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IP Allocation Table for IronOne using VLSM

Dept: / Criteria	Developers	Finance	Sales and Marketing	Customer Care	HR	Network Team	Legal
No. of Users	55	25	20	10	10	5	5
Block Size	64	32	32	16	16	8	8
Network Address	192.168.10.0	192.168.10.64	192.168.10.96	192.168.10.128	192.168.10.144	192.168.10.160	192.168.10.168
Default Gateway	192.168.10.1	192.168.10.65	192.168.10.97	192.168.10.129	192.168.10.145	192.168.10.161	192.168.10.169
First Usable IP	192.168.10.1	192.168.10.65	192.168.10.97	192.168.10.129	192.168.10.145	192.168.10.161	192.168.10.169
Subnet Mask	255.255.255.192	255.255.255.224	255.255.255.224	255.255.255.240	255.255.255.240	255.255.255.248	255.255.255.248
CIDR	/26	/27	/27	/28	/28	/29	/29
Last Usable IP	192.168.10.62	192.168.10.94	192.168.10.126	192.168.10.142	192.168.10.158	192.168.10.166	192.168.10.174
Broadcast Address	192.168.10.63	192.168.10.95	192.168.10.127	192.168.10.143	192.168.10.159	192.168.10.167	192.168.10.175
VLAN ID	10	20	30	40	50	60	70

Table 1: IP Allocation Table

192 = 11000000 255.255.255.0 = /24

255.255.255.0 = 11111111.11111111.11111111.00000000 = /24

255.255.255.192 = 11111111.11111111.11111111.11000000 = /26 =128+64=192 = developer

255.255.255.224 = 11111111.11111111.11111111.11100000 = /27 = 128+64+32 = 224 finance

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Assumptions:

1. IPv4 of IP addressing is used for above
2. No. of users in each department are sorted from highest to lowest and according to that block sizes are identified

   For example; In developers dept; 55 users are there, and the block size is calculated as follows;

   $2^n > 55$
3. Network Address is calculated by adding block size to the given IP address,

   For example: In developers dept; Network address is 192.168.10.0

   192.168.10.0 + 64 = 192.168.10.64
4. Used the First Usable IP address as the Default Gateway address assuming;

   Default Gateway = First Usable IP address

   but used the next IP address after the gateway address for PC Configuration.
5. Subnet mask and CIDR are calculated as follows;

   Partially filled Subnet Mask = 256 – Block size

   Ex: In developers dept; 192=256-64

   CIDR Value = Binary Value of Partially filled Subnet Mask

   Ex: In above example; 192 = 11000000

   Since this is a class C network; 24 network bits can be identified Adding two 1 bits to the 24 gives the CIDR Notation of the above network, which is /26 in this example
6. Last Usable Address is defined by deducting 2 IP addresses from the Network address of the next department
7. Broadcast Address is defined by deducting 1 IP address from the Network address of the next department

Default: 255.255.255.0

New SNM: 256-64 = 192 – Developer

255.255.255.192

New SNM: 256-32 = 224 – finance 255.255.255.224

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Justification

The Proposed IP allocation plan uses VLANs and subnetting using VLSM to achieve network segregation as required by the management. Each department is assigned a separate subnet, ensuring that they are isolated from each other which provides security, improves network performance, and enables easier management of network traffic.

3. Proposed Network design

Before implementation, I have designed the below blueprint by using the following devices. Assumptions are made to design the blueprint

- One 1841 Router
- *Six 2950-24 Switches
    - One for Server Room
    - Five for Departments
- *20 PCs
- *2 Server-PTs
- Copper Straight-Through cables
    - For connecting router to switch to servers in server room
    - For connecting router to switch in developer department
    - For connecting PCs to Switch in each department
- Copper Cross-over Cables
    - For connecting switches among departments

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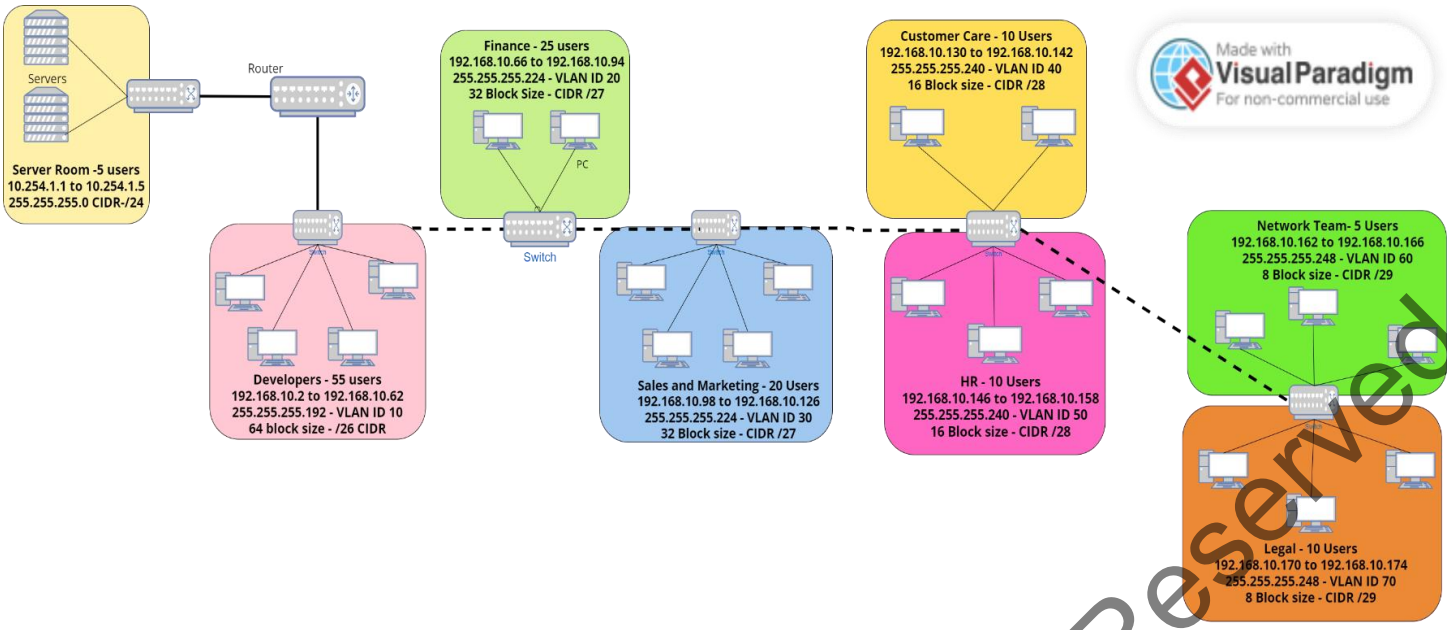


Figure 3- 1: Blueprint for IronOne Network

Assumptions

1. *To build this network, we use Ethernet based 2950-24 LAN Switches. Since each switch is in 24 ports, we need below no. of switches for each department according to their requirement.

Department	No. of Users	No. of Switches
Developers	55	3
Finance	25	1
Sales and Marketing	20	1
Customer Care	10	1
HR	10	1
Network Team	5	1

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Legal	5	1
	130	9

2. Since this is the proposed plan, I have used 2 PCs to represent each department and assigned first usable IP and last usable IP for IP configuration of PCs. For actual implementation we need to use 130 PCs, representing users for each department respectively

3. Since we are using 2950-24 Switches having 24 ports; I have connected
    - Customer Care department and HR Department to one switch;
    - Network Team Department with Legal Department to one switch

    to simplify the network design and to reduce the no. of devices required, resulting in easier management and troubleshooting.

4. I have used 2 Server PTs representing server room and assigned first and last usable IP. No server configuration is done as it's a separate network. For actual implementation, we need to use 5 servers for server room and assumed that server room has its own dedicated switch to handle the network infrastructure within the area

Justification

Overall, this network design ensures departmental isolation, efficient local connectivity, and centralized routing functionality, providing necessary infrastructure to support the given requirements, allowing better control and management of network resources, and facilitates the future growth and scalability of the network.

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4. Implementation of the Proposed Network Design

There are several software tools available for implementing and managing network infrastructure.

1. **Cisco Packet Tracer:** Cisco Packet Tracer is a network simulation and visualization tool widely used for network design, configuration, and troubleshooting.

2. **Graphical Network Simulator-3(GNS3)** is an open-source network emulation software that allows users to build, configure, and test complex network topologies.

3. **Cisco Network Assistant:** Cisco Network Assistant is a graphical network management tool designed for Cisco network devices.

4. **SolarWinds Network Configuration Manager:** SolarWinds Network Configuration Manager is a comprehensive network management software that provides configuration and change management for network devices.

The choice of the best software depends on the specific needs, complexity of the network, and the organization's preferences. Cisco Packet Tracer and GNS3 are popular choices for network simulation and design, especially in educational and learning environments. Cisco Network Assistant is suitable for managing Cisco network devices, while SolarWinds Network Configuration Manager offers comprehensive network management features for larger networks.

I have used Cisco Packet Tracer 8.2.1 Version Software to implement the blueprint as

- It allows for the simulation of network behavior, enabling users to visualize how network traffic flows and troubleshoot potential issues.

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- Specifically designed for Cisco network devices, offering a comprehensive range of Cisco switches, routers, and other networking equipment. This makes it an ideal choice for simulating and configuring Cisco-based network infrastructures.
- It is available for Windows, macOS, and Linux, ensuring compatibility with various operating systems and facilitating network design and configuration on different platforms.

While other network management software like GNS3, Cisco Network Assistant, and SolarWinds Network Configuration Manager offer advanced features and broader network management capabilities, Cisco Packet Tracer is specifically designed for network simulation and configuration, making it well-suited for implementing the network design outlined in this scenario.

4.1. Building the blueprint

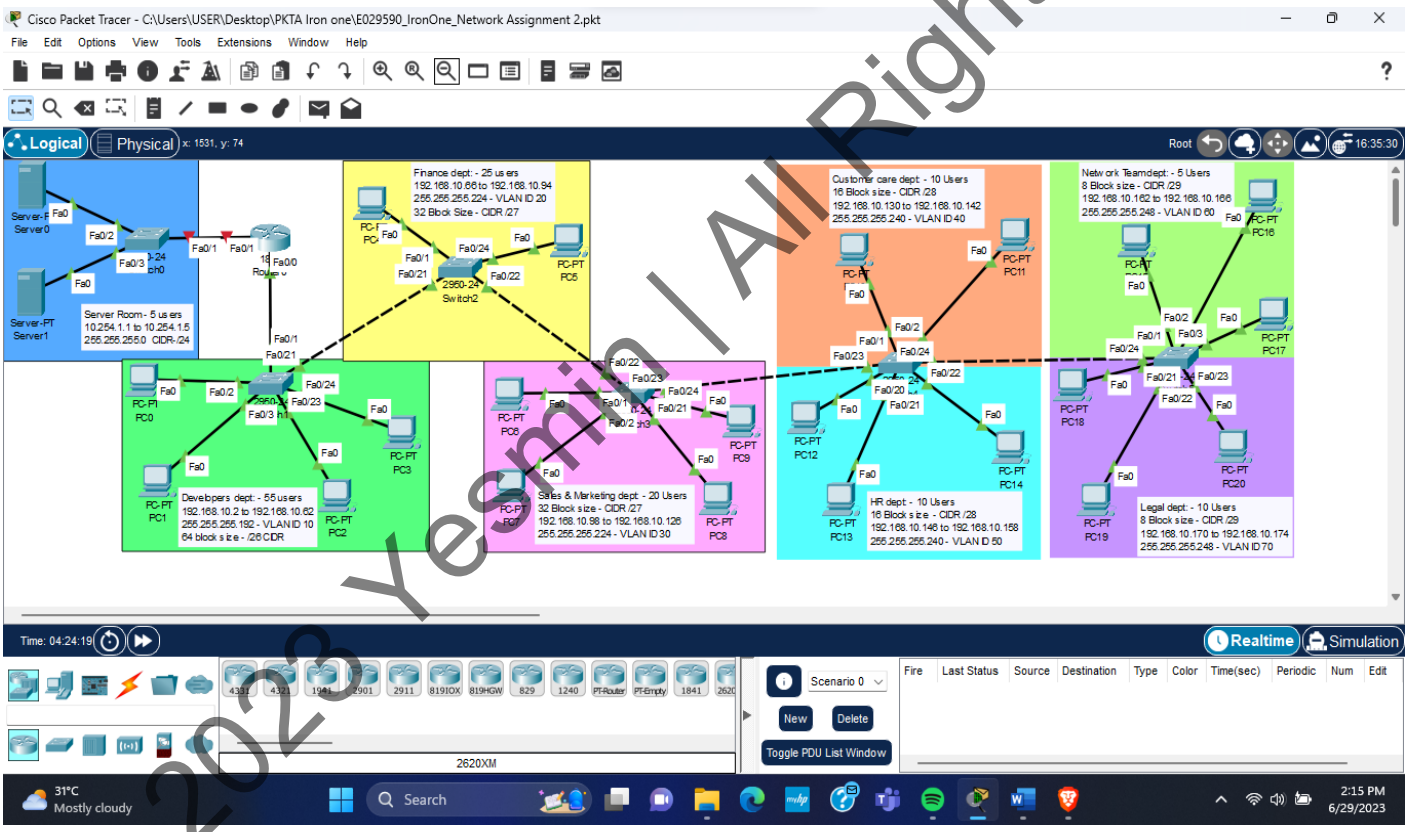


Figure 4- 1: Implementation of IronOne Network

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4.1.1. PC Configuration

In here, the Usable IP address ranges founded on the IP Allocation Table are assigned to PCs .

Note that the next usable IP address are assigned after the Gateway Address

Developer Department IP assign

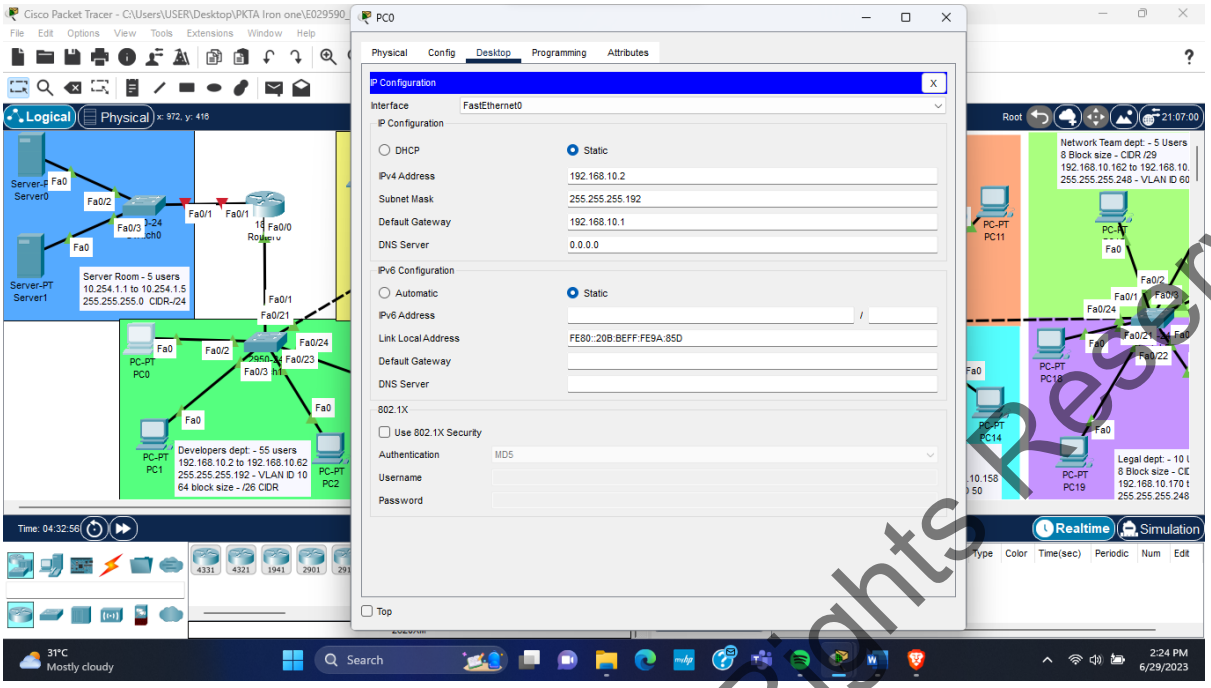


Figure 4- 2:IP assign for PC0 in Developer Department

Finance Department IP Assign

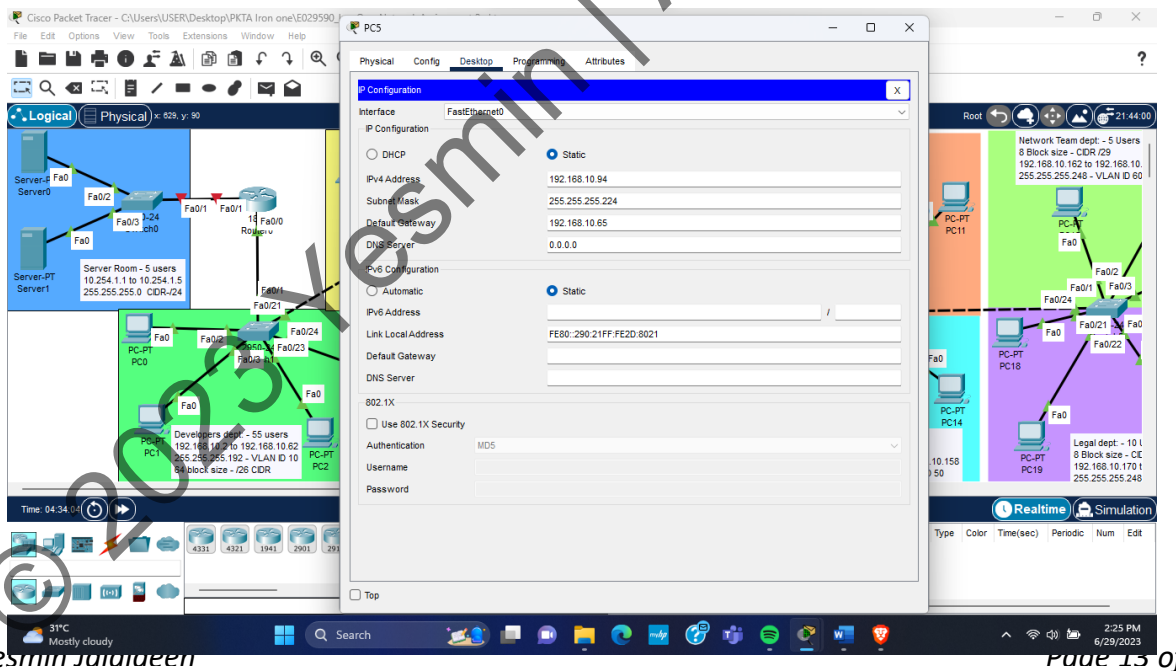


Figure 4- 3:IP assign for PC5 in Finance Department

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Sales and Marketing Department IP Assign

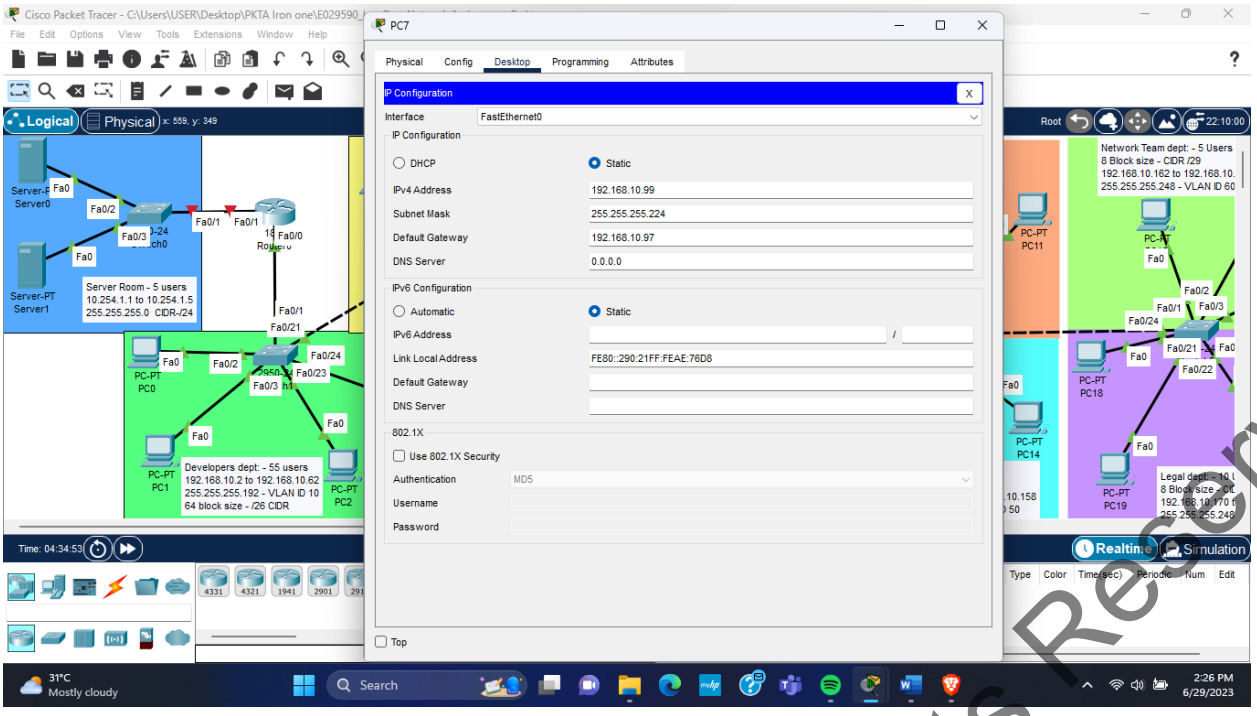


Figure 4- 4:IP assign for PC7 in Sales and Marketing Department

Customer Care Department IP Assign

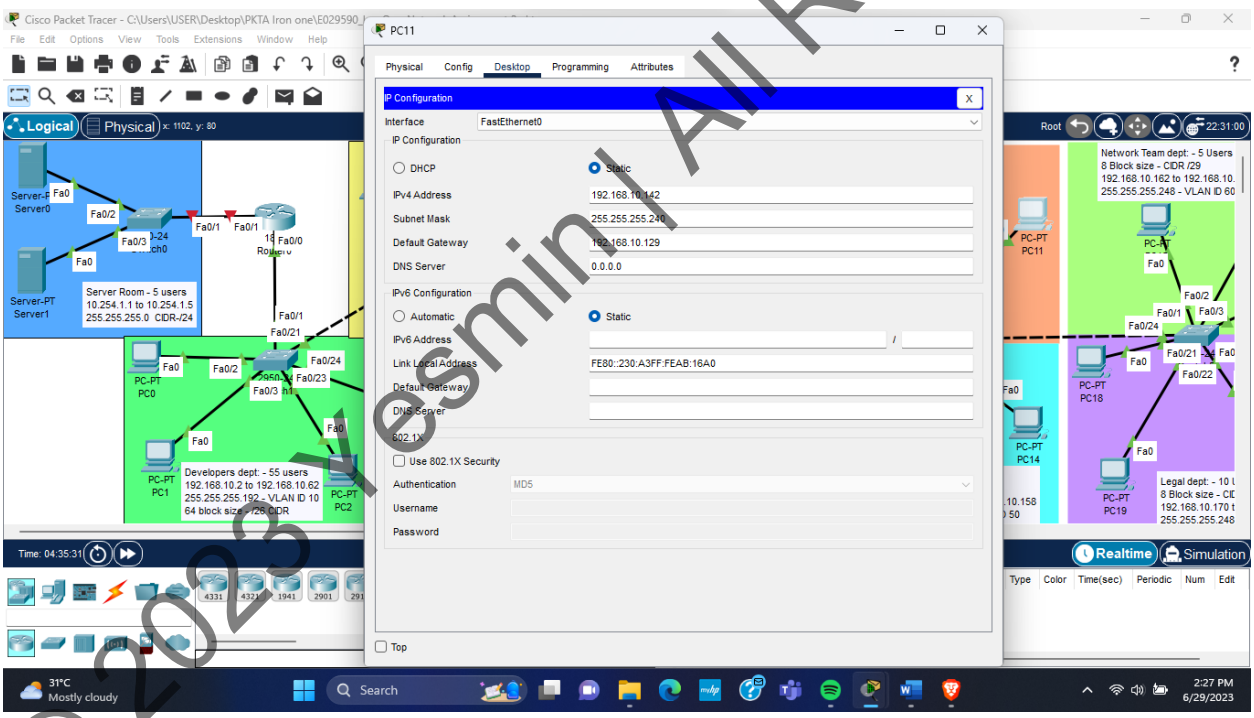


Figure 4- 6:IP assign for PC11 in Customer Care Department

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HR Department IP Assign

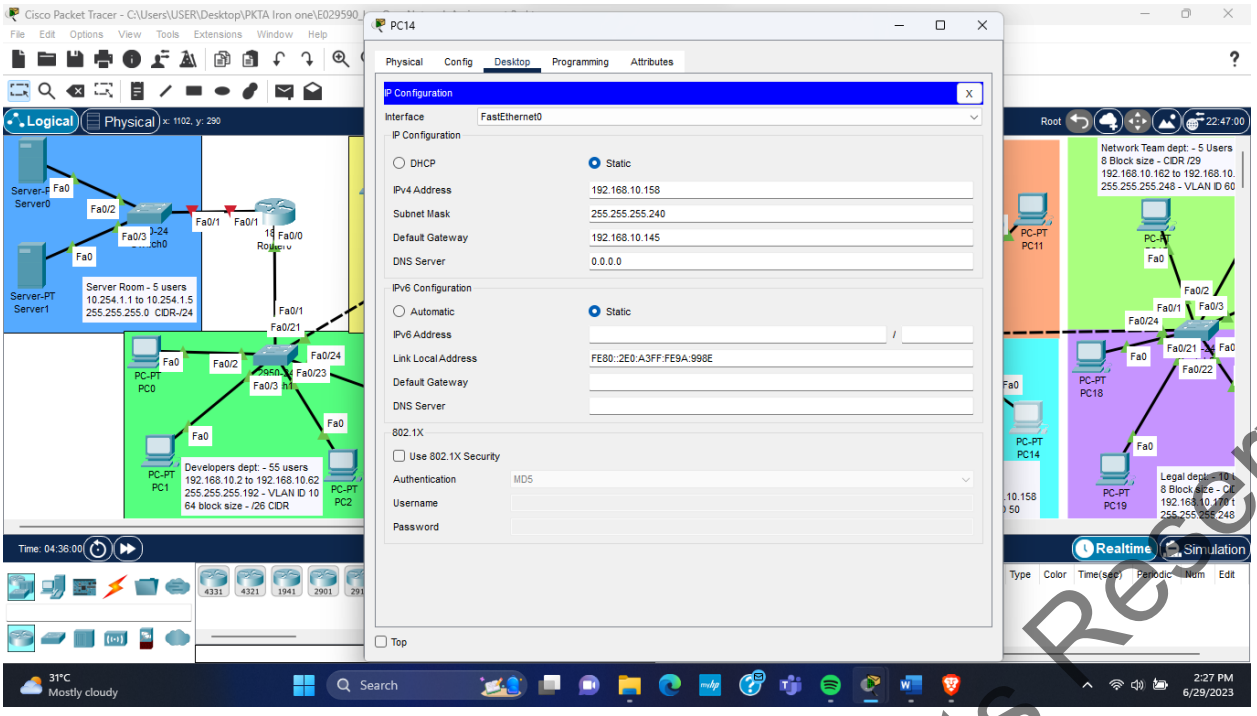


Figure 4- 8:IP assign for PC14 in HR Department

Network Team Department IP Assign

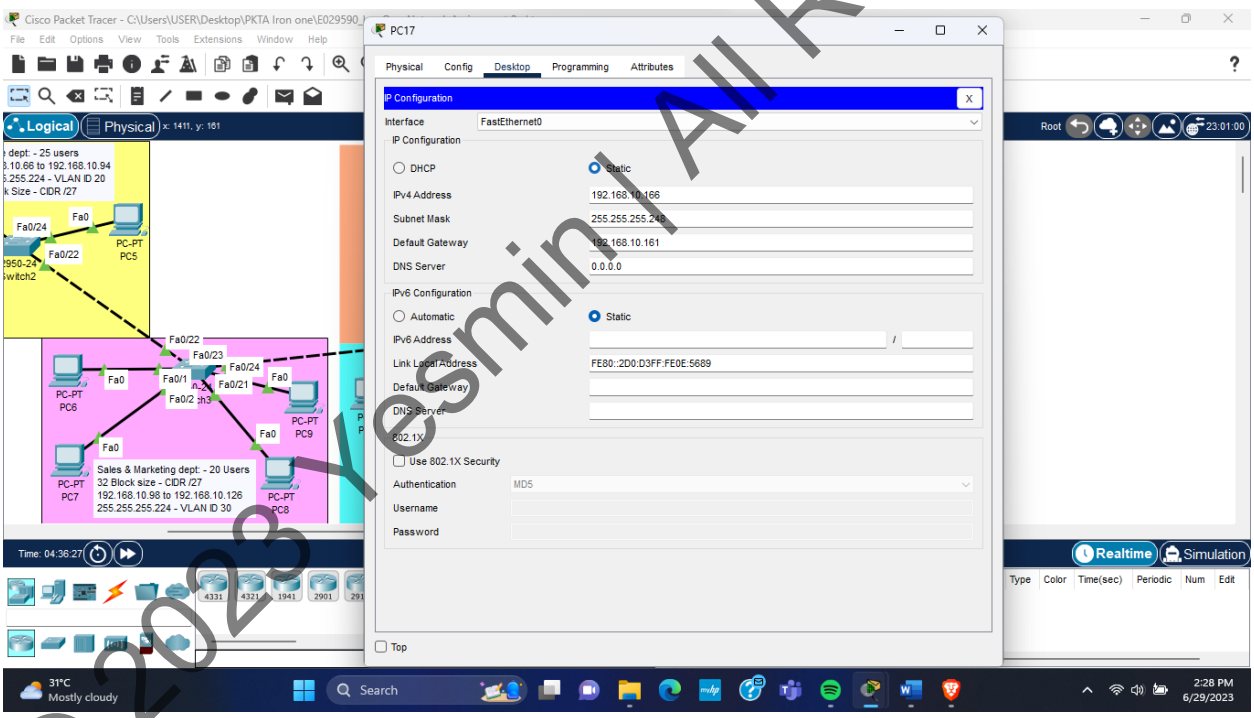


Figure 4- 9:IP assign for PC17 in Network Team Department

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Legal Department IP Assign

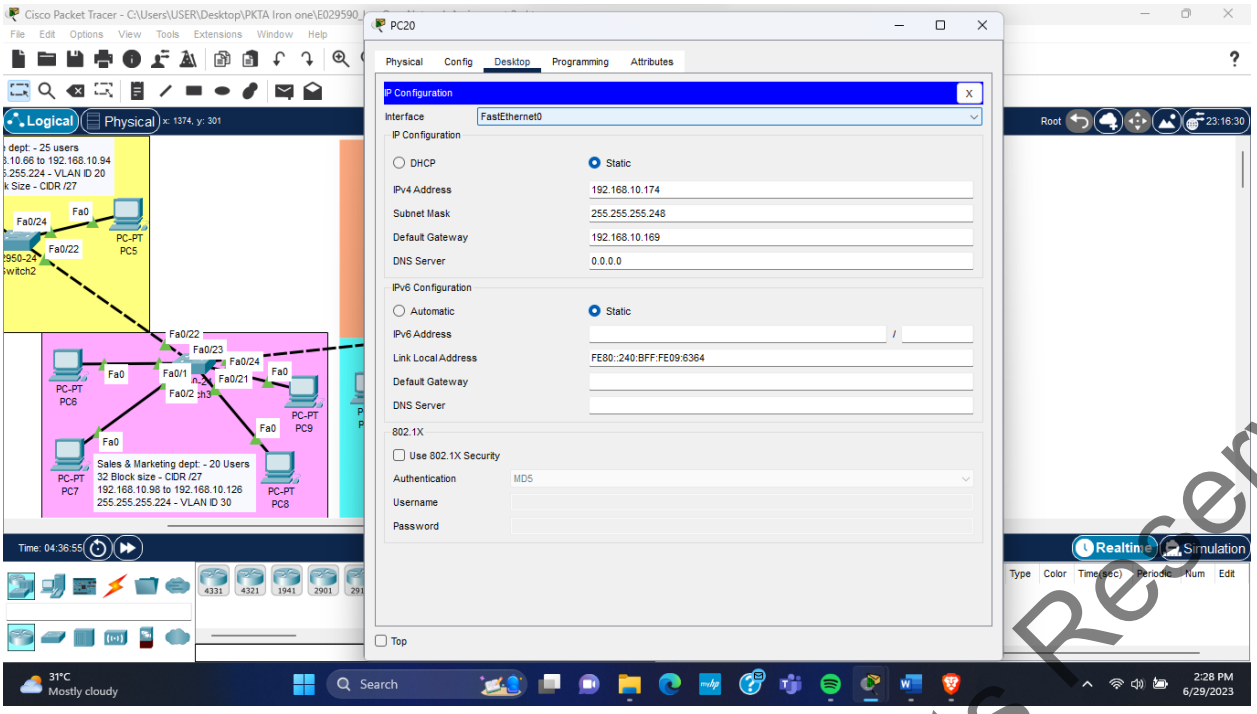


Figure 4- 10:IP assign for PC20 in Legal Department

Server Room Configuration : Only IP addresses are assigned, and no other configurations done

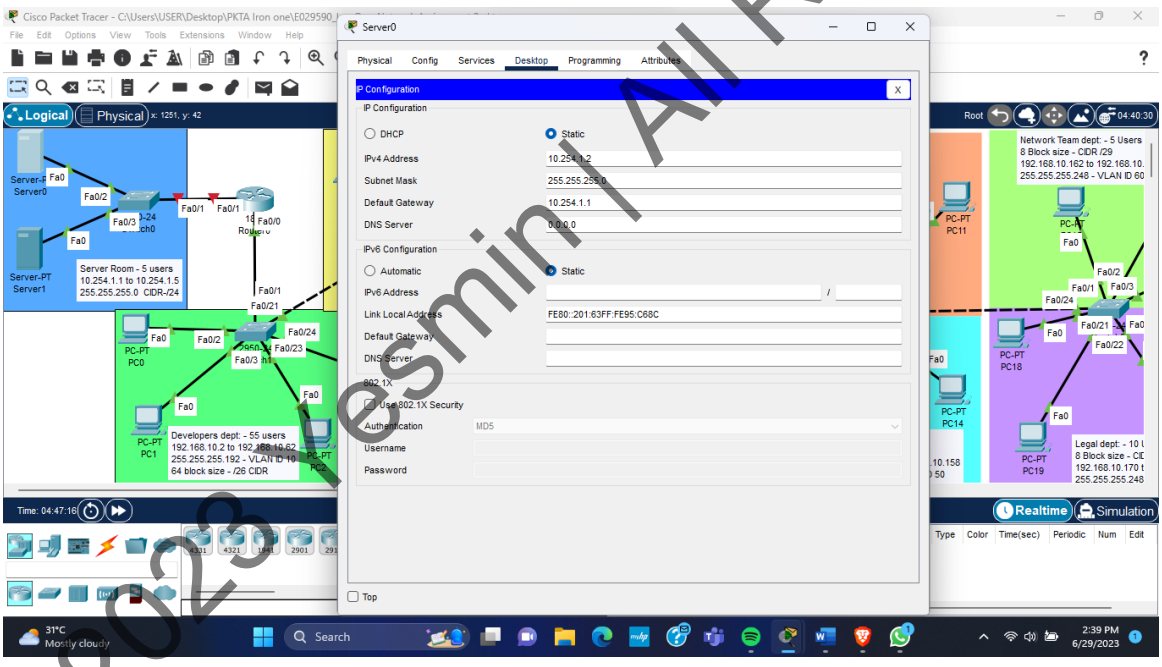


Figure 4- 11:IP assign for Server0 in Server Room

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4.1.2. VLAN Configuration for Switches

VLAN Configuration for Developer Department (SwA)

Creating and Setting VLAN : I have created a hostname and set all the VLANs in the network first.

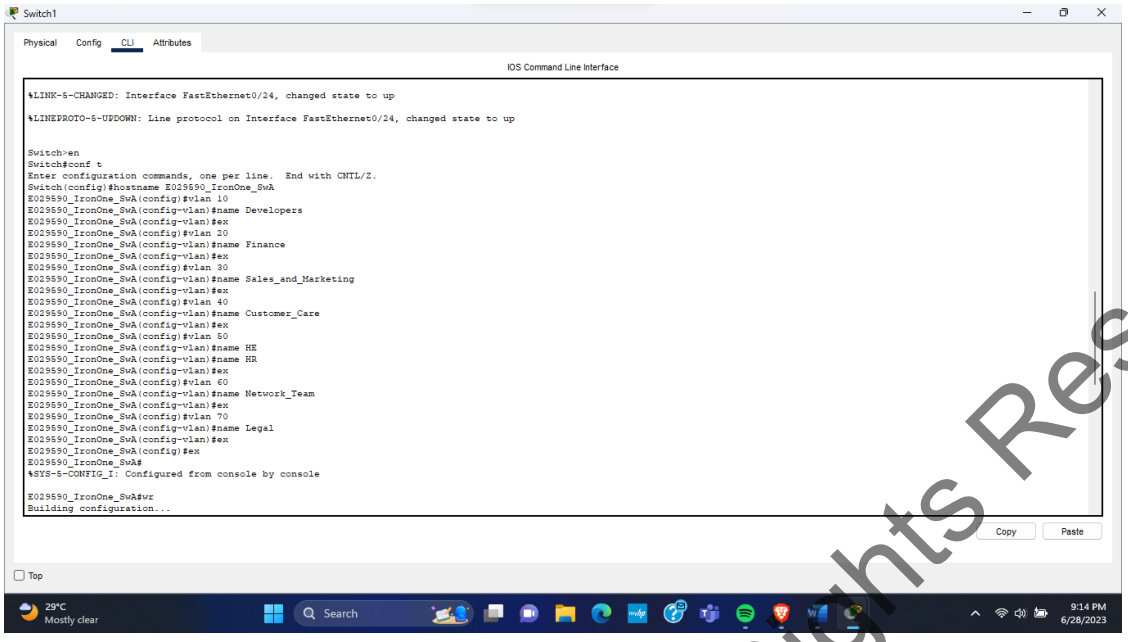


Figure 4- 12: Hostname and VLAN ID creation for Switch A

Assigning Switchports to the VLAN and Trunk-link Configuration

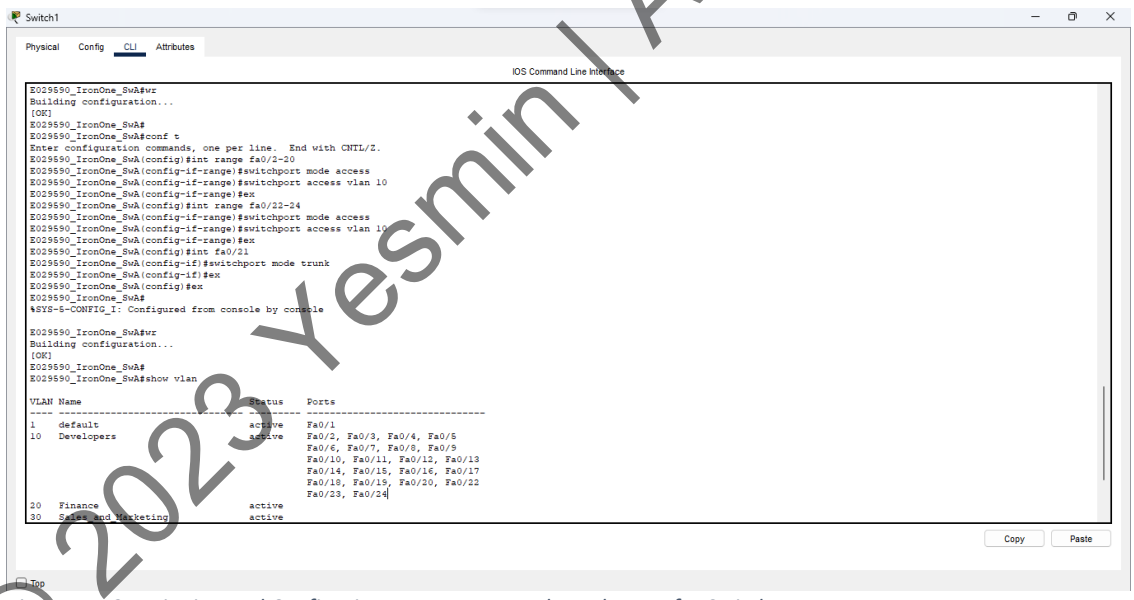


Figure 4- 13:Assigning and Configuring Access ports and Trunk ports for Switch A

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Final outcome of VLAN by using Show VLAN Command

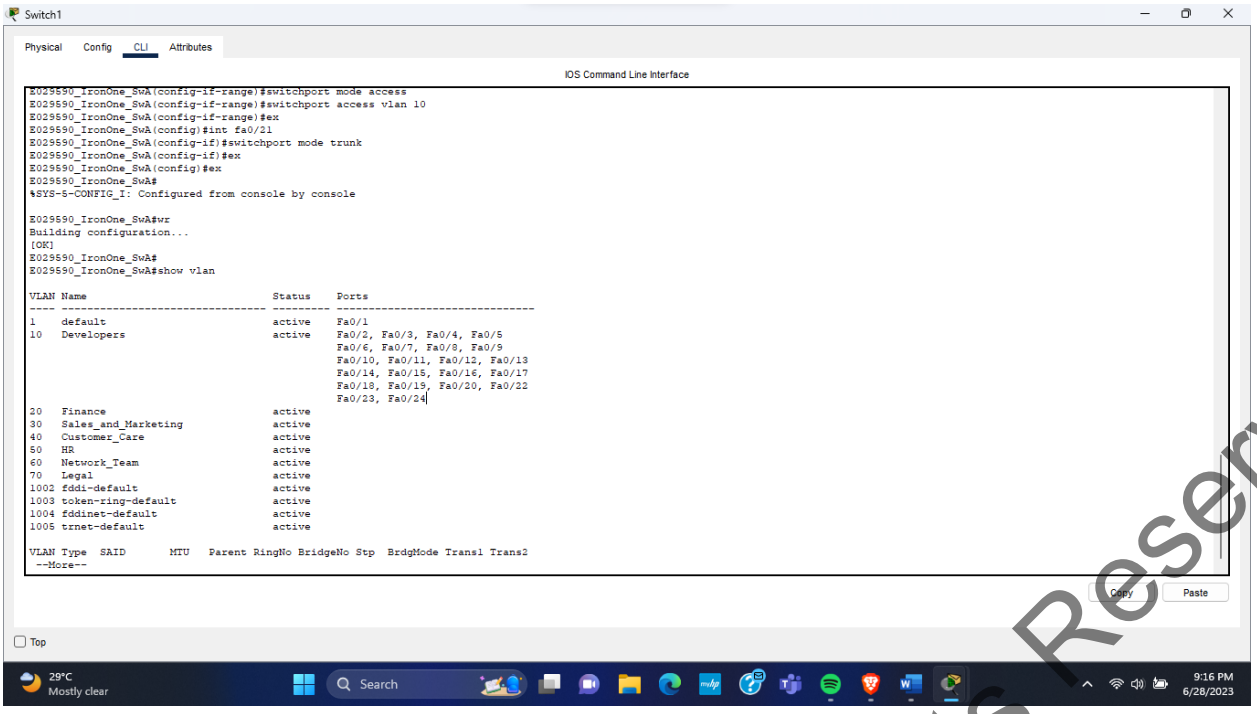


Figure 4- 14:VLAN Outcome using show VLAN command in Switch A

VLAN Configuration for other departments are done as follows; (Screenshots are attached by the above order)

VLAN Configuration for Finance Department (SwB)

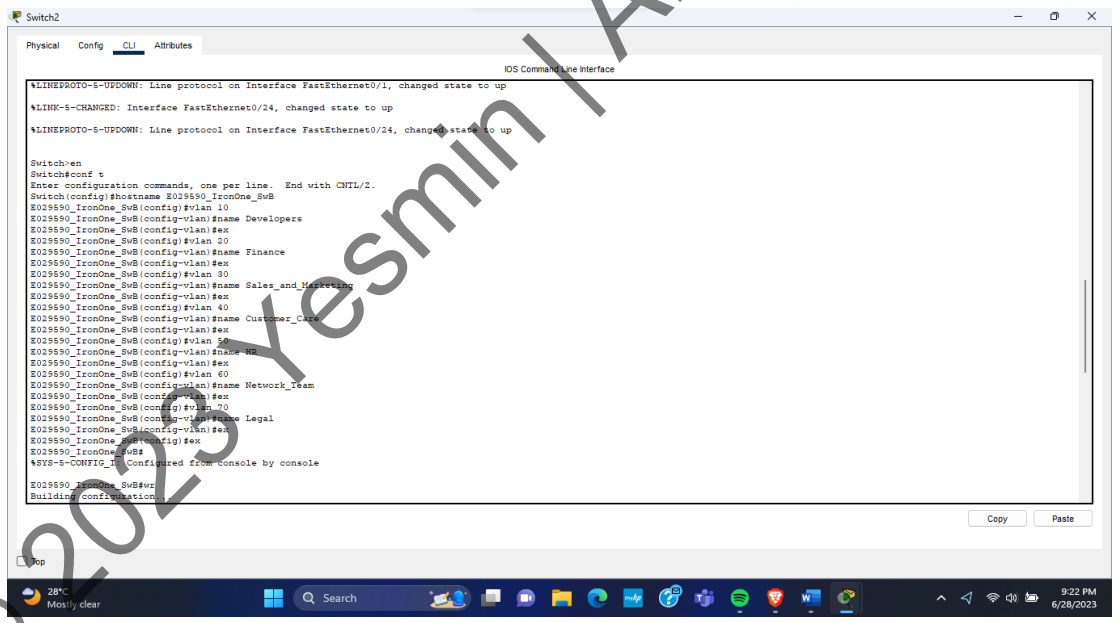


Figure 4- 15:Hostname and VLAN ID creation for Switch B

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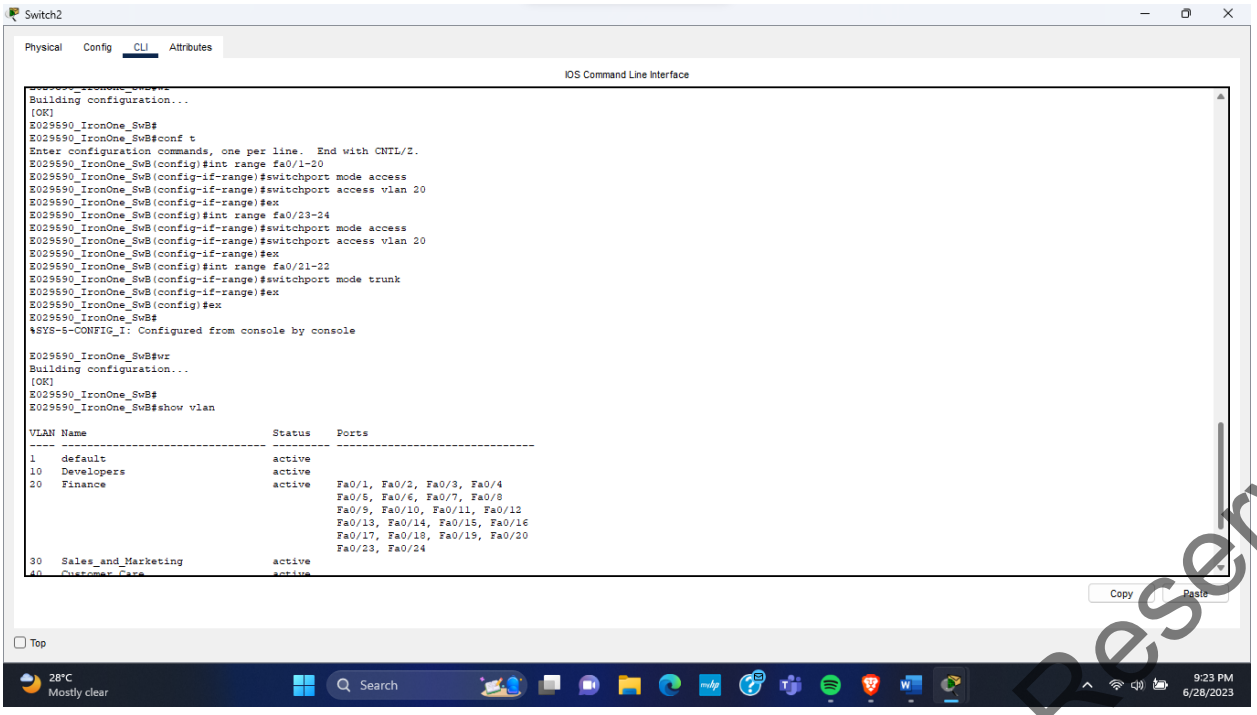


Figure 4- 16:Assigning and Configuring Access ports and Trunk ports for Switch B

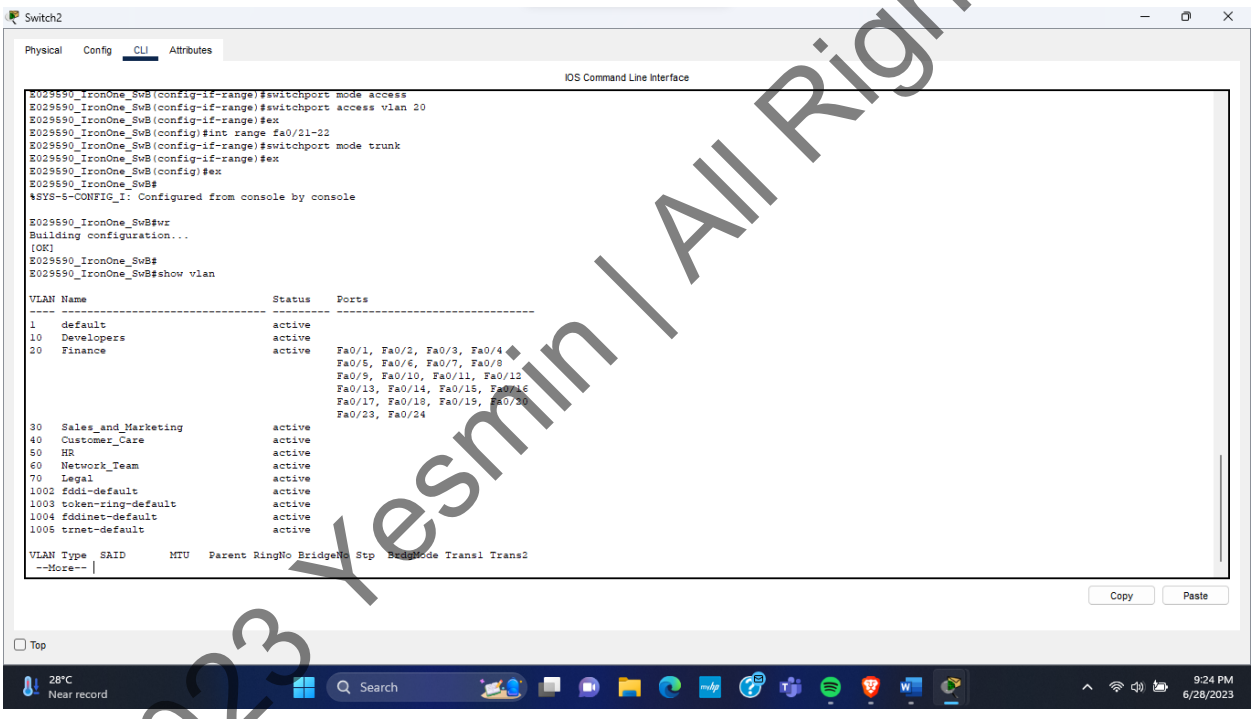


Figure 4- 17:VLAN Outcome using show VLAN command in Switch B

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VLAN Configuration for Sales and Marketing Department (SwC)

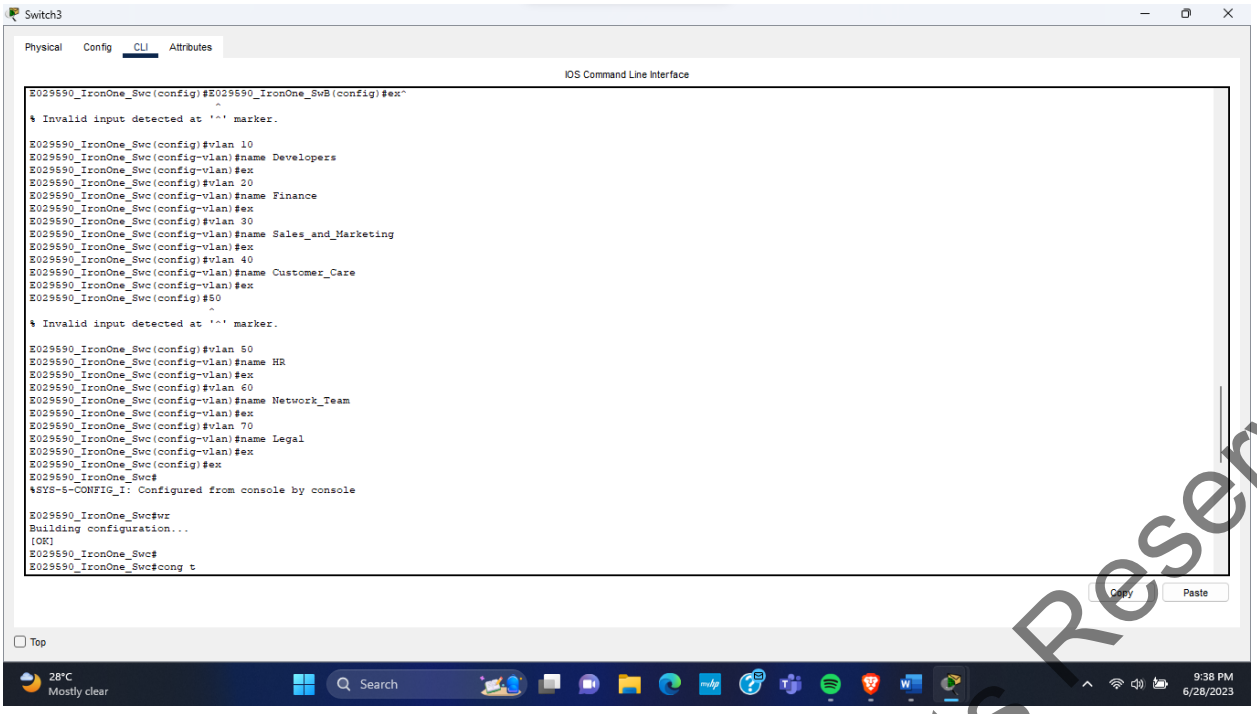


Figure 4- 18:Hostname and VLAN ID creation for Switch C

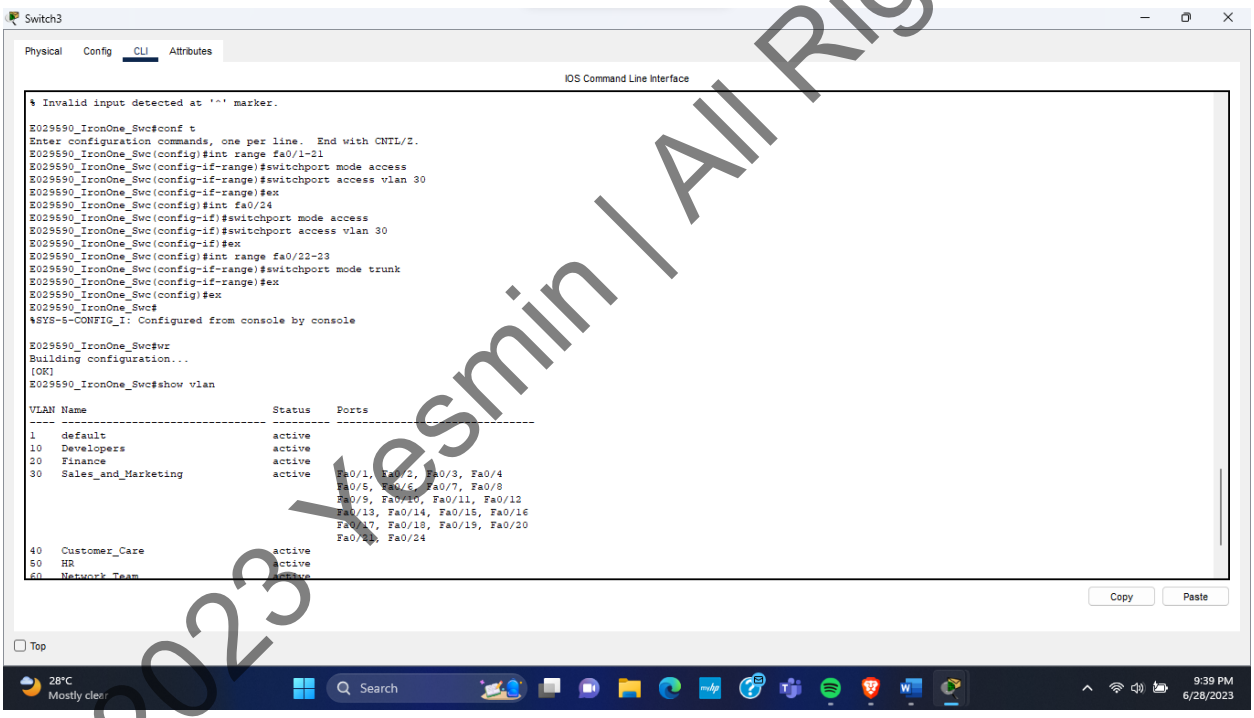


Figure 4- 19:Assigning and Configuring Access ports and Trunk ports for Switch C and VLAN Outcome

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VLAN Configuration for Customer Care Department and HR Department (SwD)

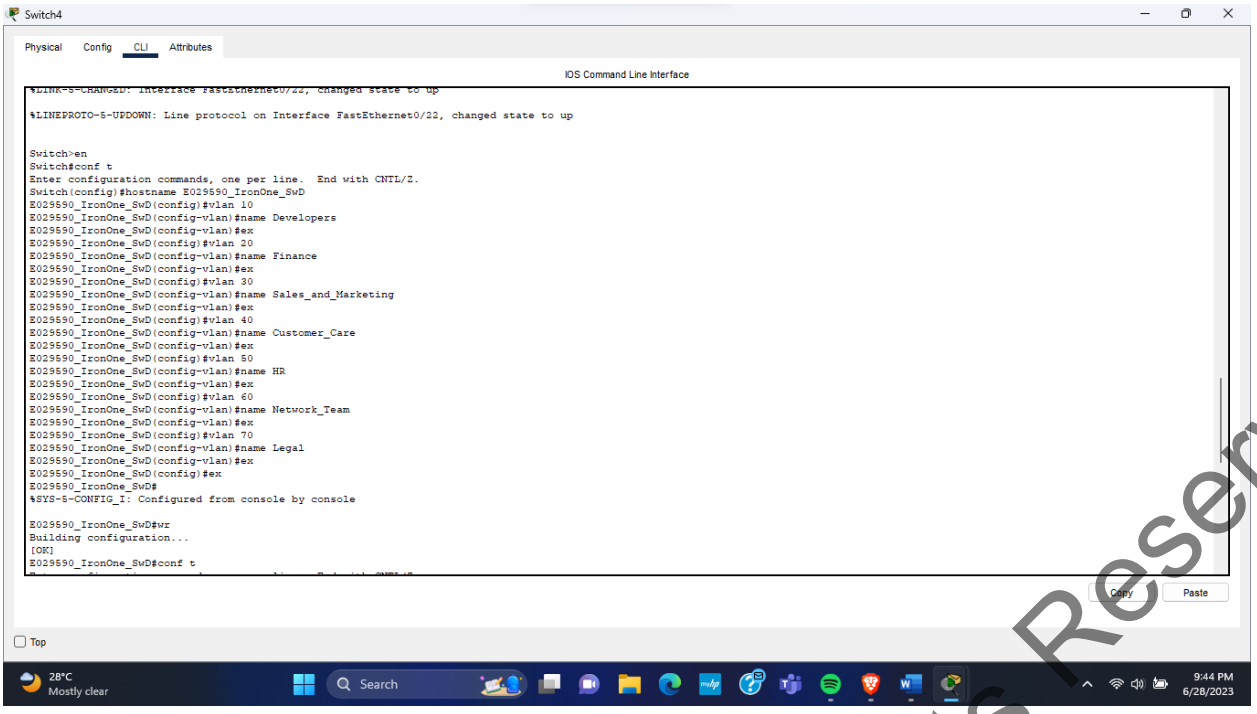


Figure 4- 20:Hostname and VLAN ID creation for Switch D

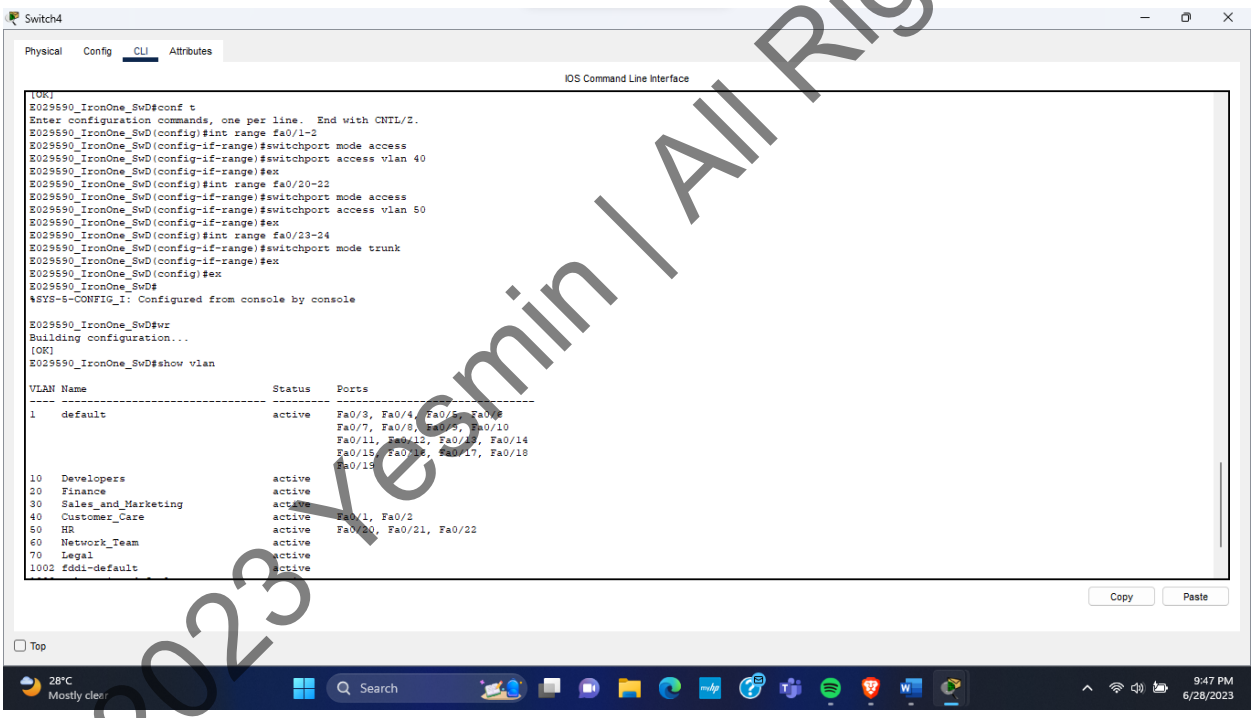


Figure 4- 21:Assigning and Configuring Access ports and Trunk ports for Switch D and VLAN Outcome

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VLAN Configuration for Network Team and Legal Department (SwE)

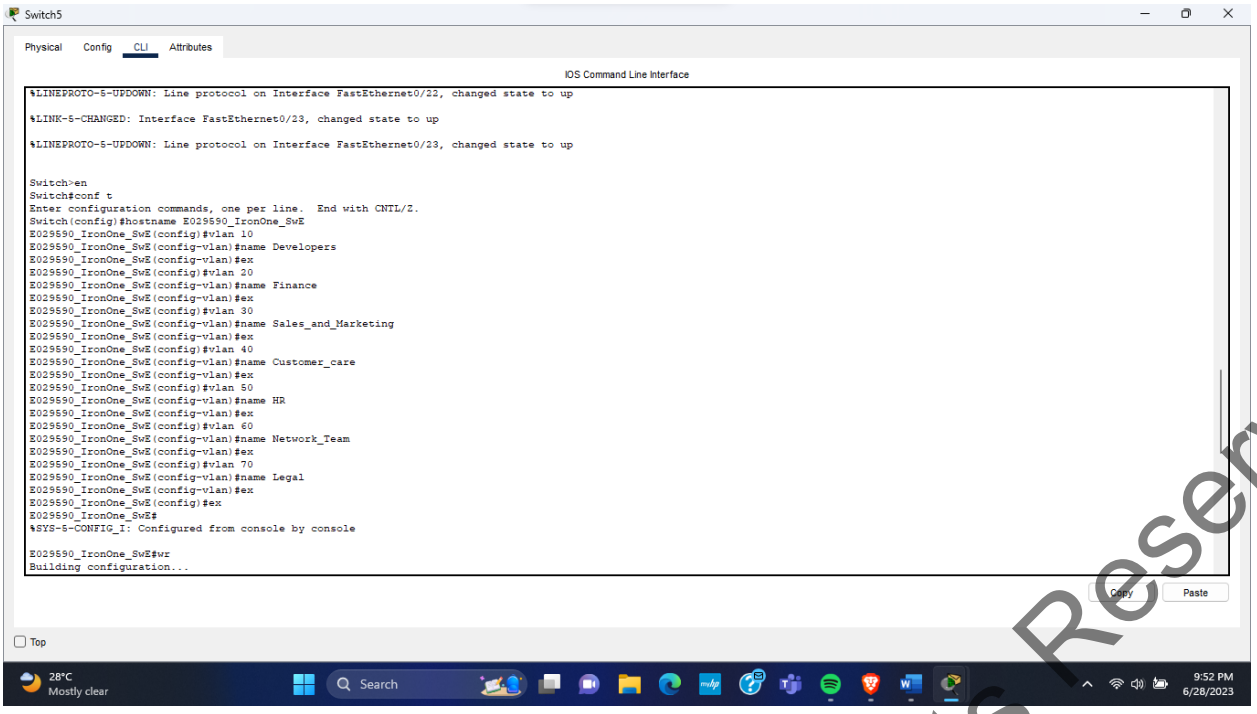


Figure 4- 22:Hostname and VLAN ID creation for Switch E

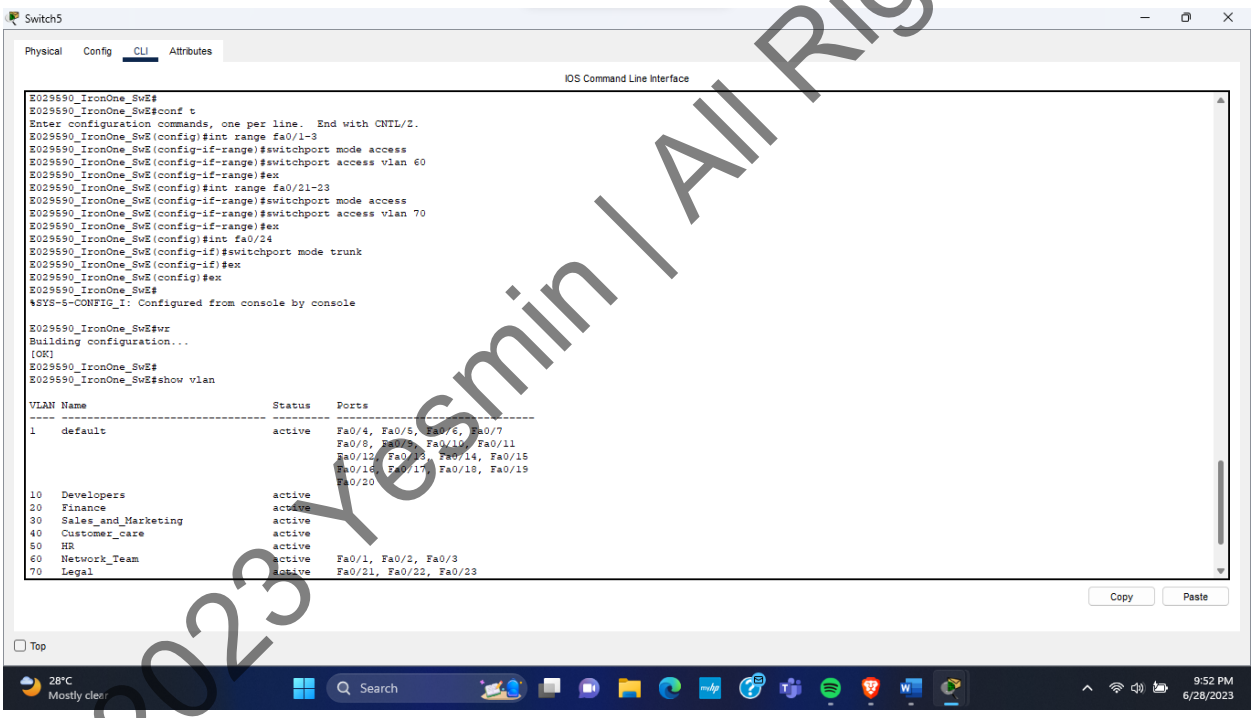


Figure 4- 23:Assigning and Configuring Access ports and Trunk ports for Switch E and VLAN Outcome

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4.1.3. Router Configuration

4.1.3.1. Creating Hostname Banner and Setting Clock

In below figure 4-22, I have given the commands for creating hostname banner and set clock and here's the final outcome of the below as Figure 4-23



Figure 4- 24: Creating Hostname Banner and Set the Clock of the Router

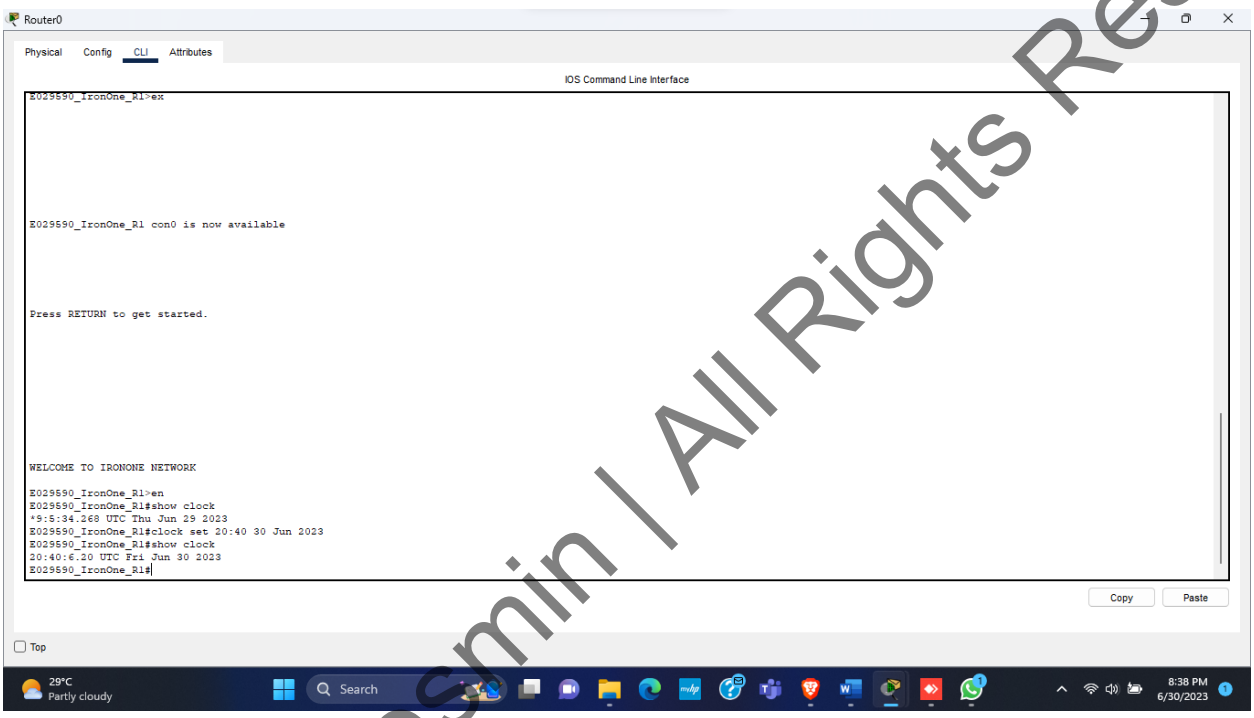


Figure 4- 25: Final Outcome

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4.1.3.2. Making the main router interface up

For this part I didn't give an IP address for Fa0/0 port and didn't up the Fa0/1 port in server room first but then I gave IP addresses to both ports to verify test cases.

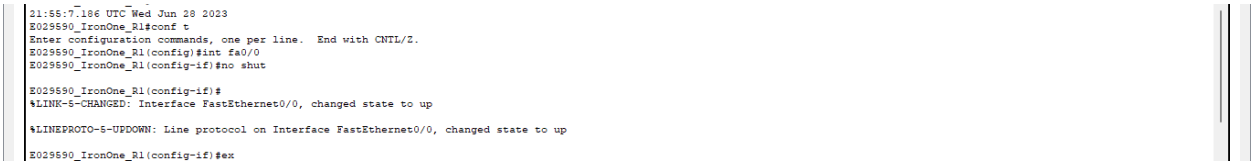


Figure 4- 26: Fa0/0 port configuration without IP address

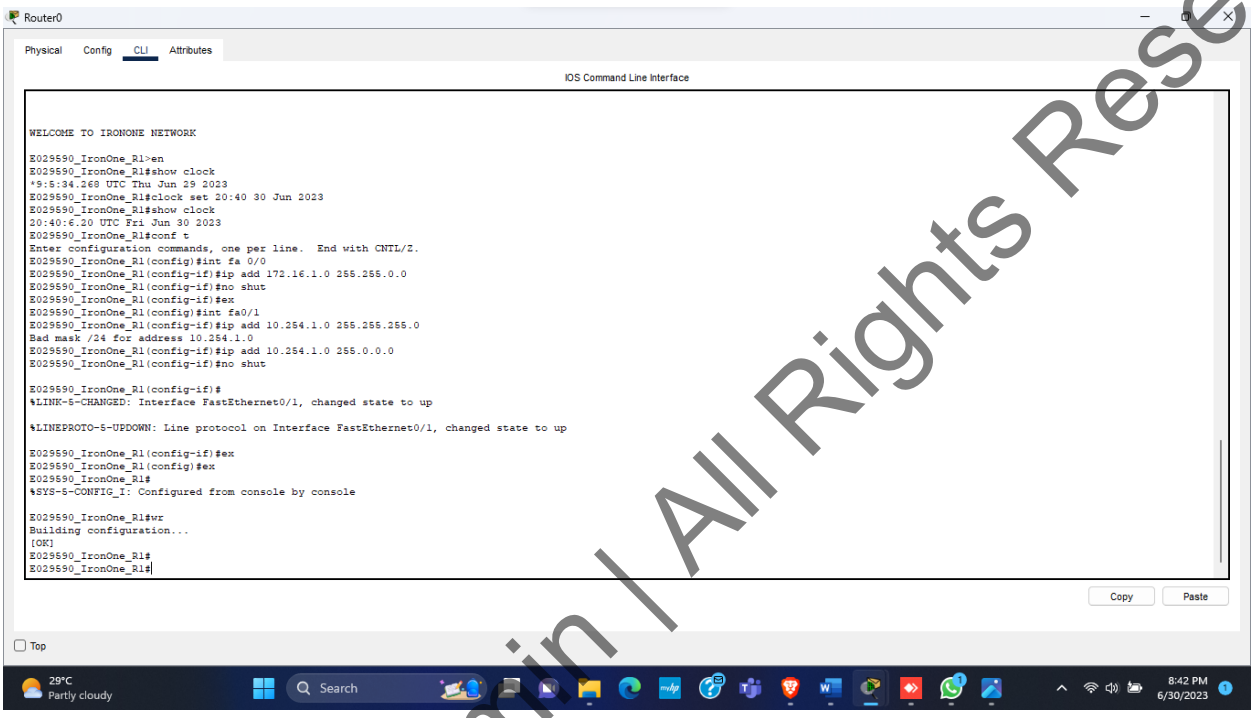


Figure 4- 27: Fa0/0 and Fa0/1 Port Configuration with IP addresses

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4.1.3.3. Creating Router sub interfaces

In this part I have already configured Fa0/0.1 while creating hostname banner and setting the clock in figure 4-22 and I have uploaded the entire configuration of the figure 4.22

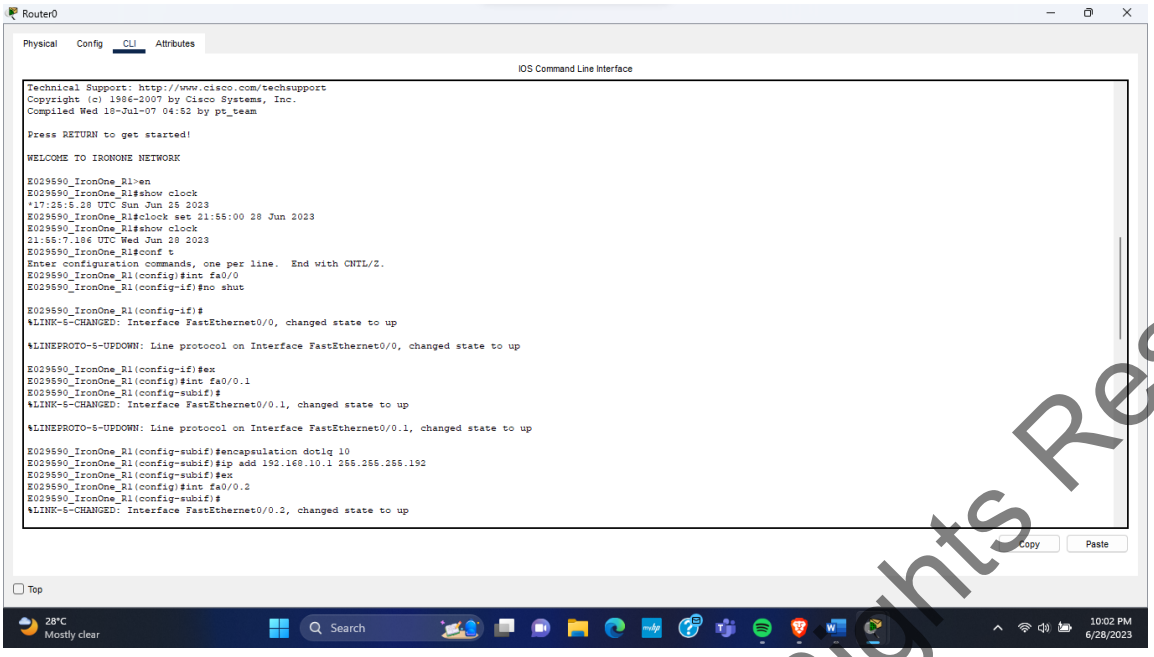


Figure 4- 28: Router Sub Interfaces 1

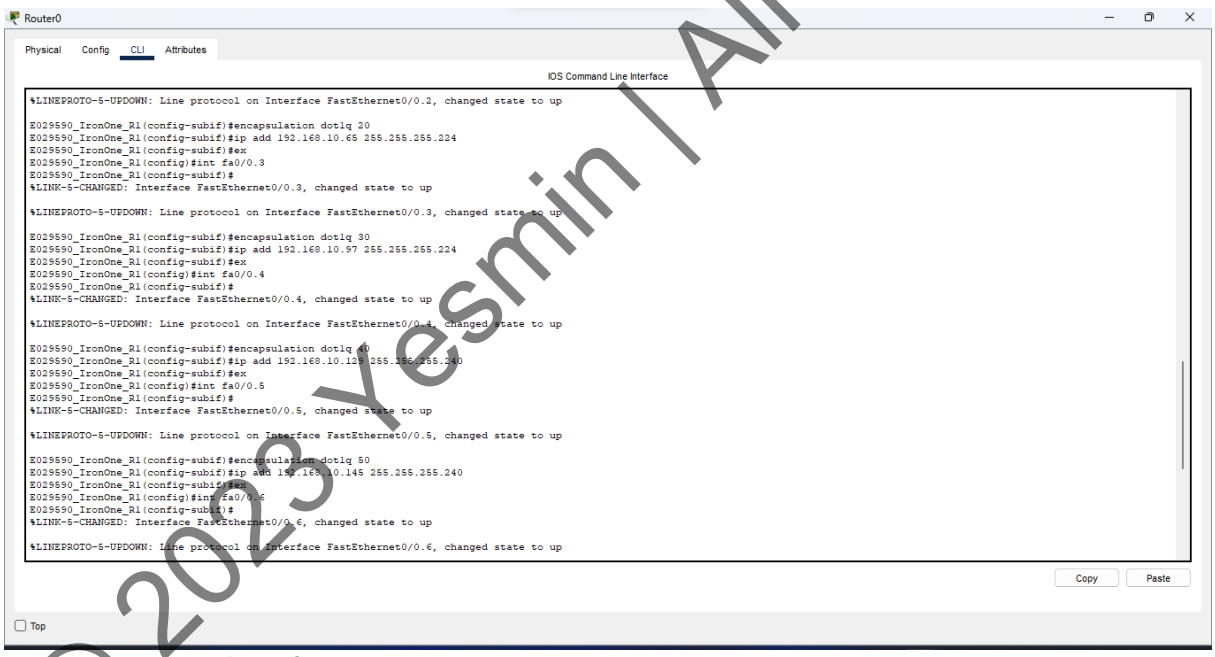


Figure 4- 29: Router Sub Interfaces 2

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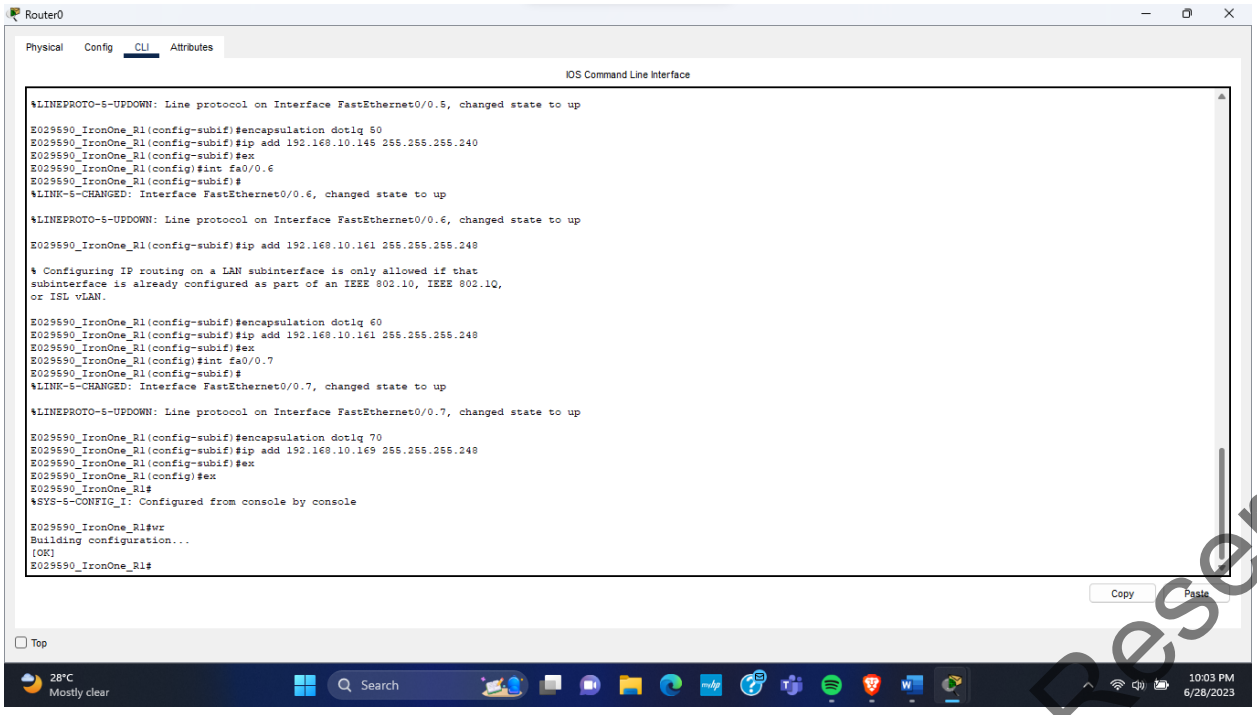


Figure 4- 30: : Router Sub Interfaces 3

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5. Testing the Implemented Network Design

Testing the network involves conducting various tests to validate the functionality, performance, and security of the implemented network. It is important to perform testing to ensure that the network meets the desired requirements and functions as intended.

By performing test cases, we can

1. Verify Network Functionality as these checks whether the network design has been implemented correctly and that devices can connect
2. Identify and Resolve issues as administrators can pinpoint potential problems and take corrective actions to improve the network's performance and reliability.
3. Validate Security measures as this will identify vulnerabilities, unauthorized access points, or potential security risks and strengthen network security and protect sensitive data from unauthorized access or breaches.
4. Performance evaluation as administrators can evaluate the performance of the network infrastructure, including factors such as network latency, bandwidth utilization, and throughput and identify any limitations or bottlenecks that may impact the network's ability to handle the expected workload or scale to accommodate future growth.
5. Enhance Compatibility between different devices, protocols, and applications within the network
6.

For the testing in the above scenario, I have done the following testing

5.1. Ping Test

Perform ping tests between devices in different VLANs to verify connectivity and isolation. This can be done in two ways.

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Ping between PCs within the same department

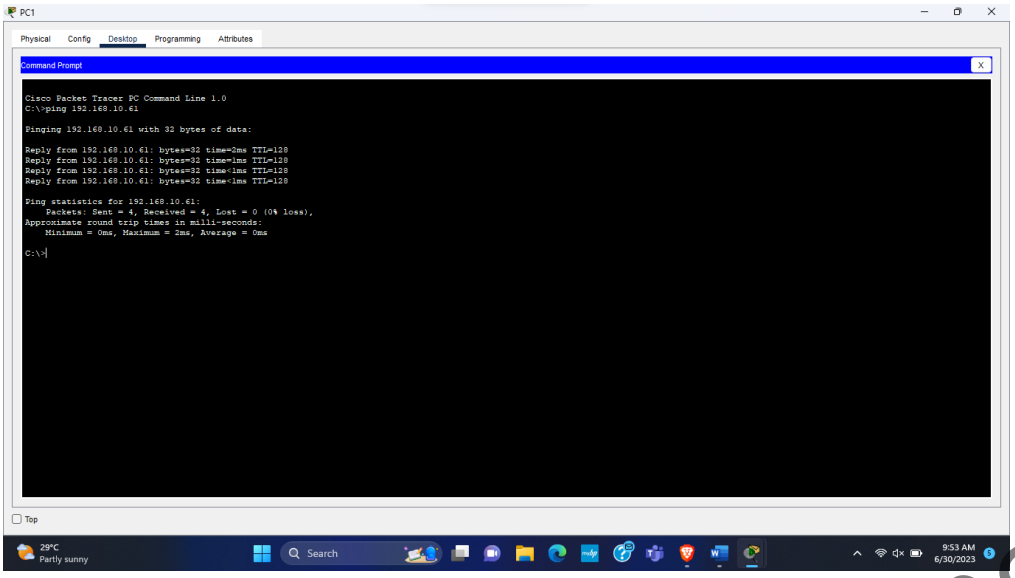


Figure 5- 1: Developer PC1 to Developer PC2

Ping between PCs in different departments

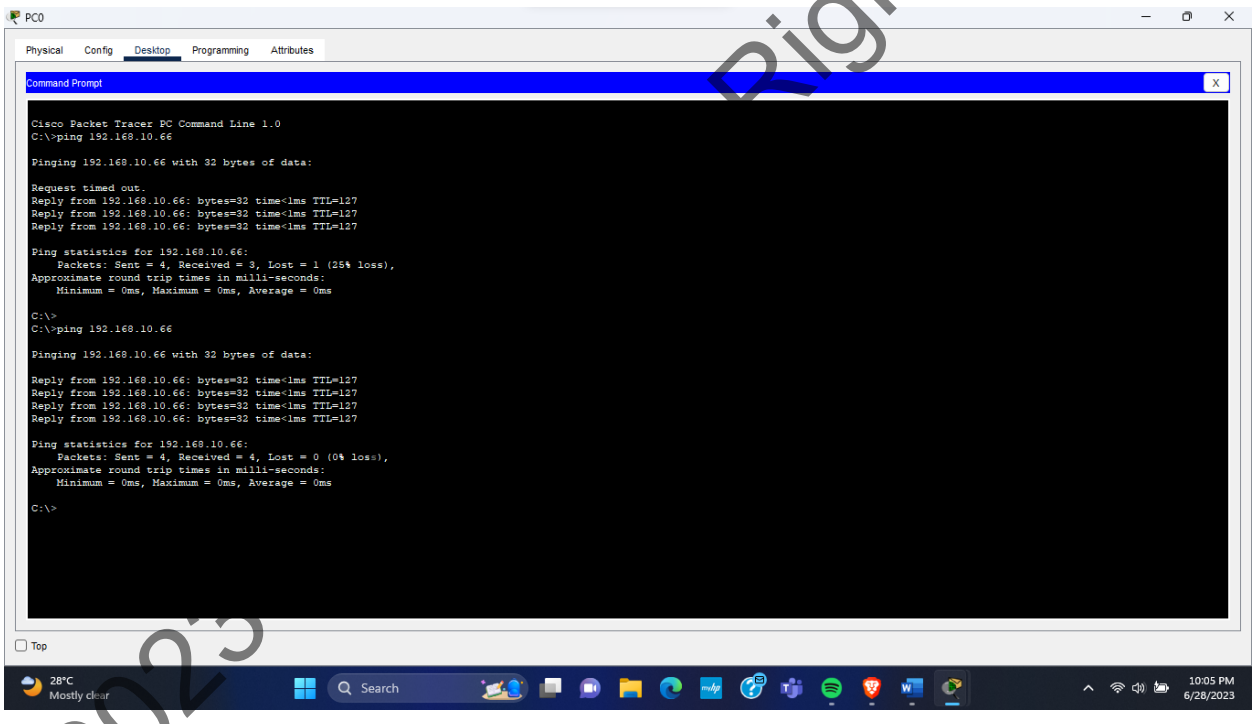


Figure 5- 2:Developer PC0 to Finance PC4

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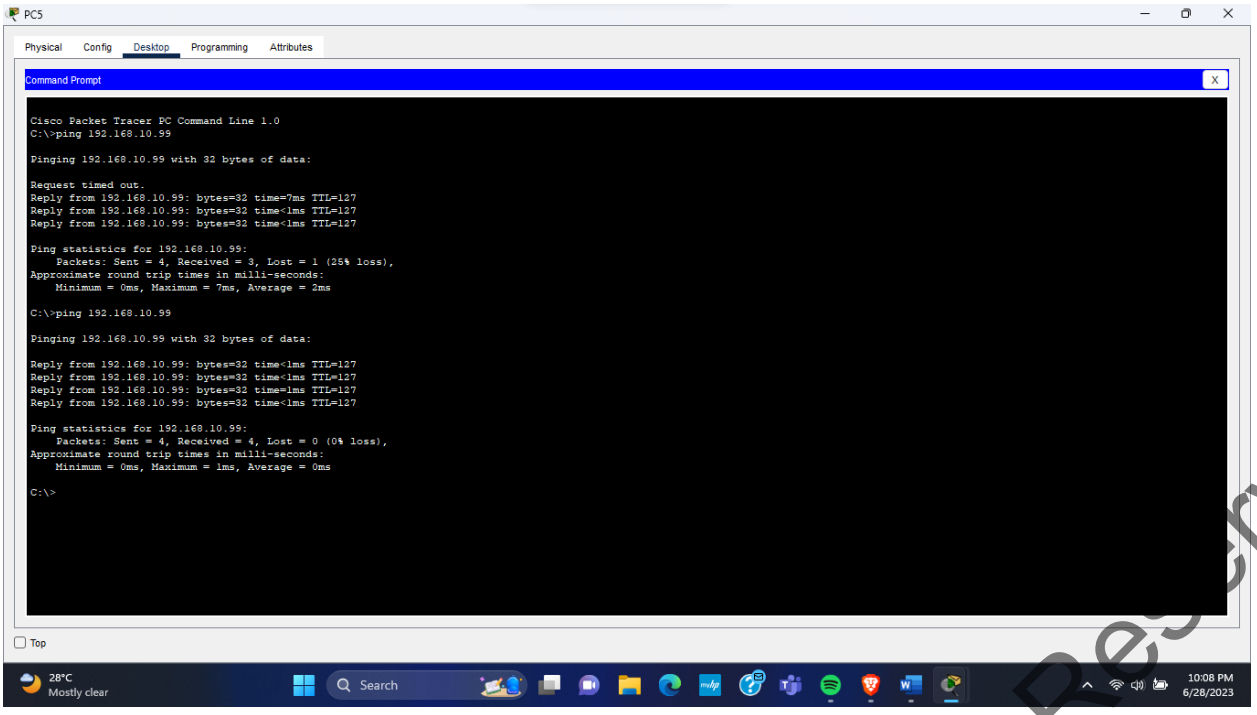


Figure 5- 3:Finance PC5 to Sales PC7

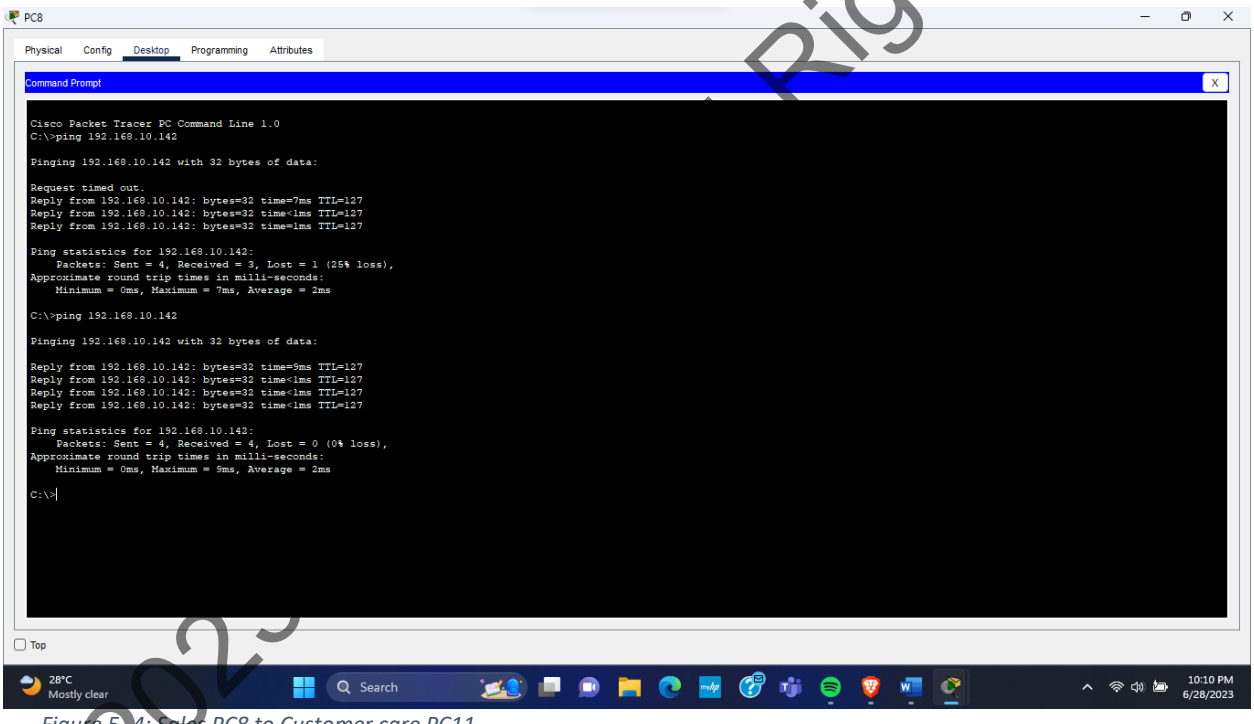


Figure 5- 4: Sales PC8 to Customer care PC11

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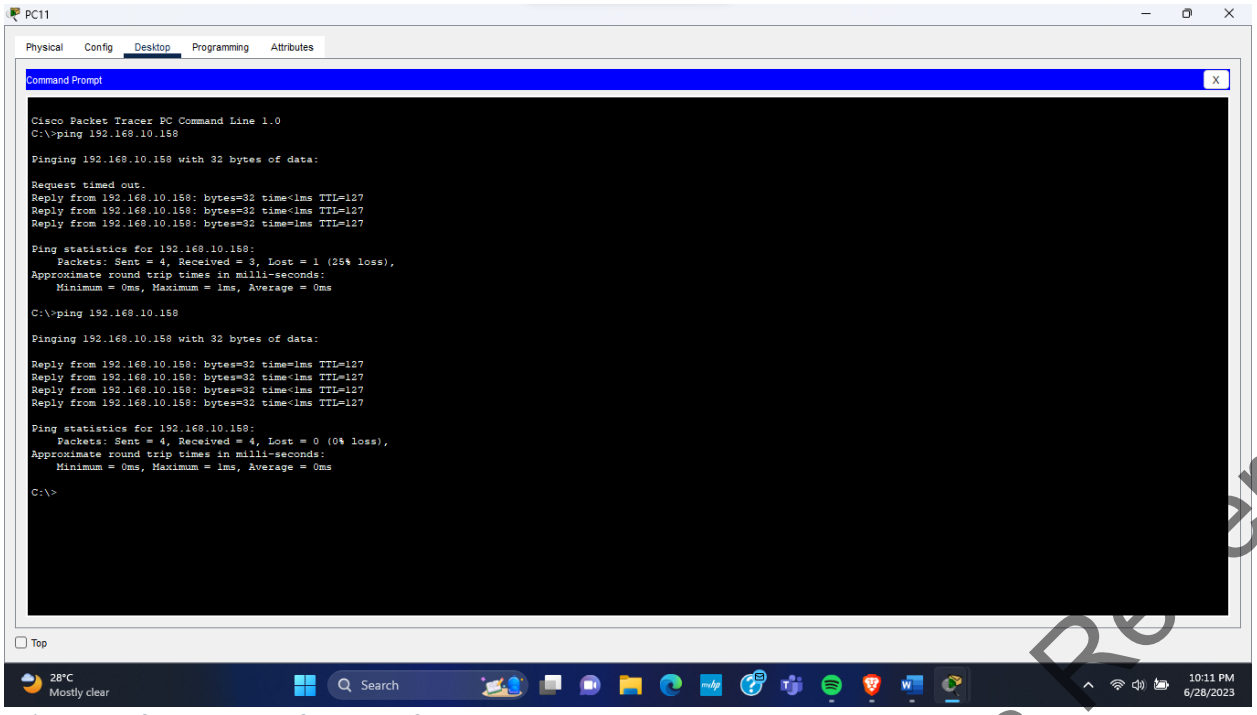


Figure 5- 5: Customer care PC11 to HR PC14

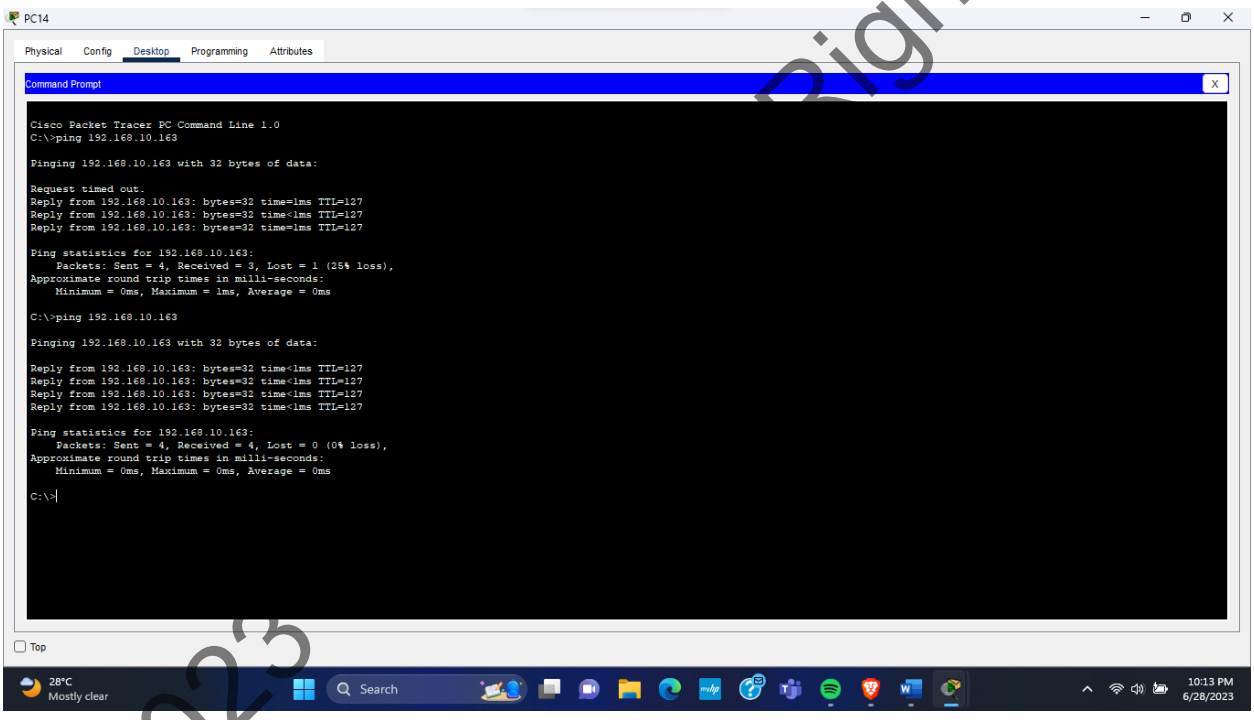


Figure 5- 6: HR PC14 to Net Team PC16

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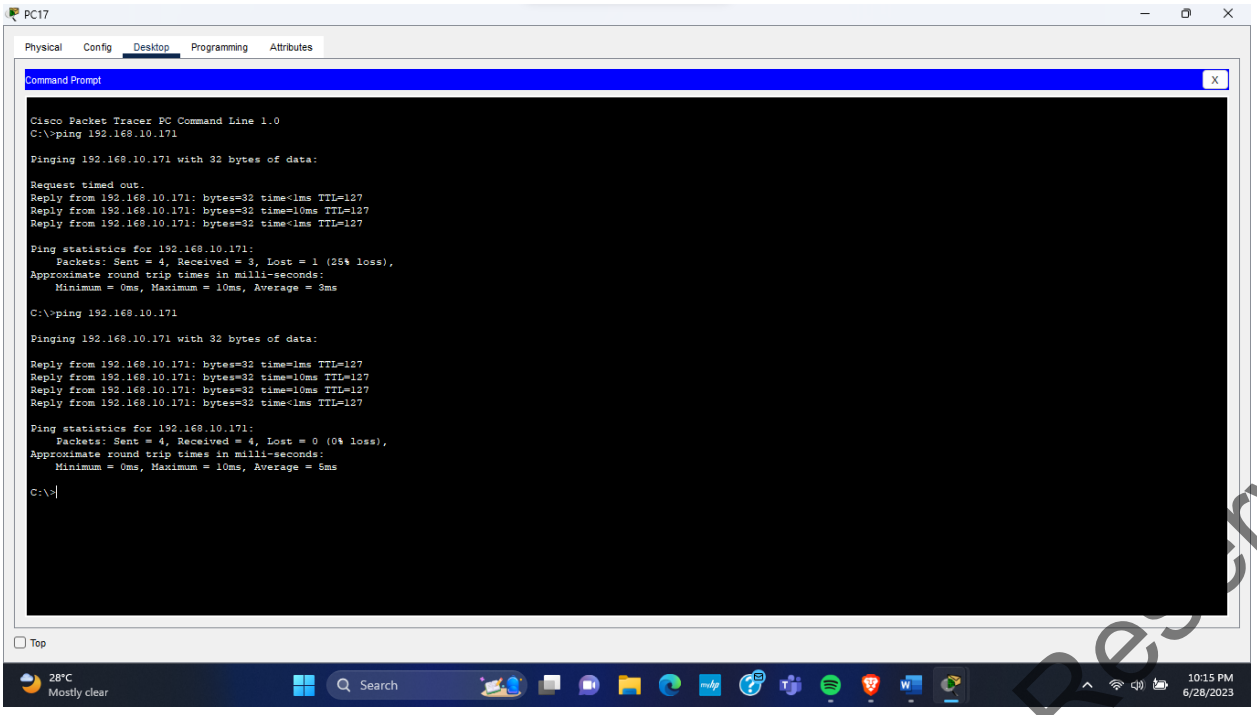


Figure 5- 7: Net team PC17 to Legal PC19

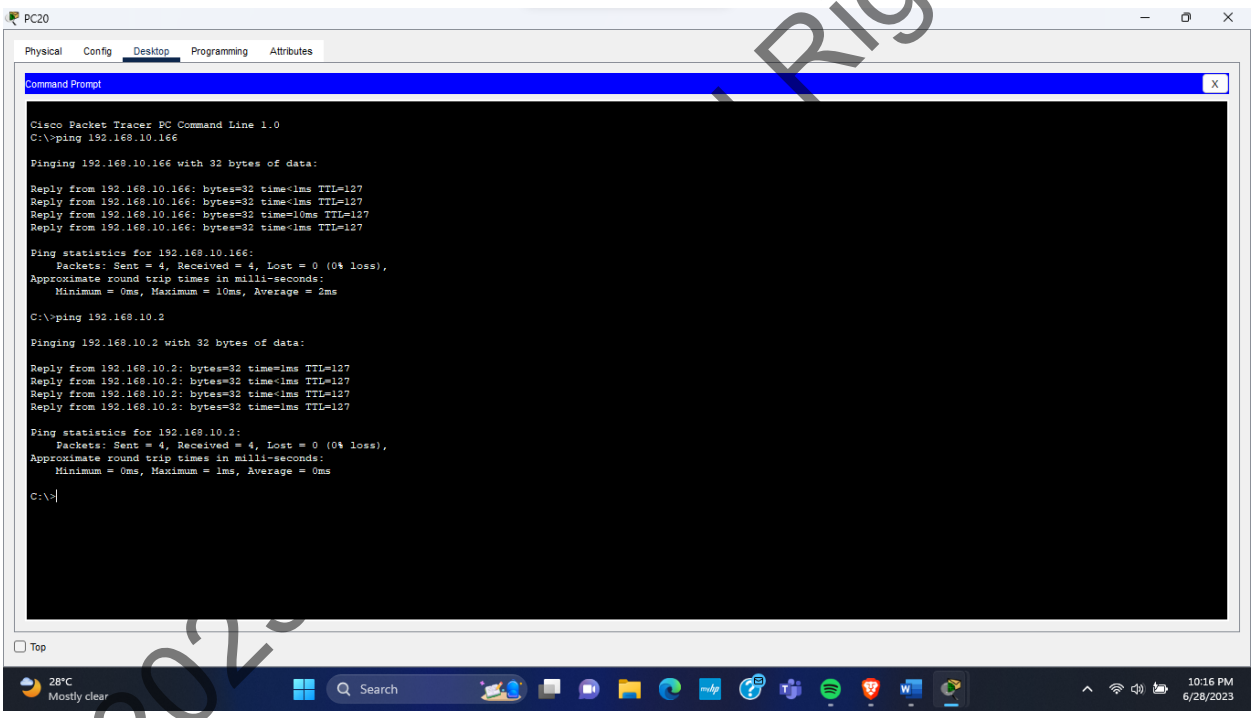


Figure 5- 8: Legal PC20 to Net team PC17 and Developer PC0

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Assumption: Since I have done the IP assign only in the server room, No configuration done in switch in the server room and no ping test done to server

5.2. Ipconfig /all

Used to retrieve comprehensive information about the network configuration of a device

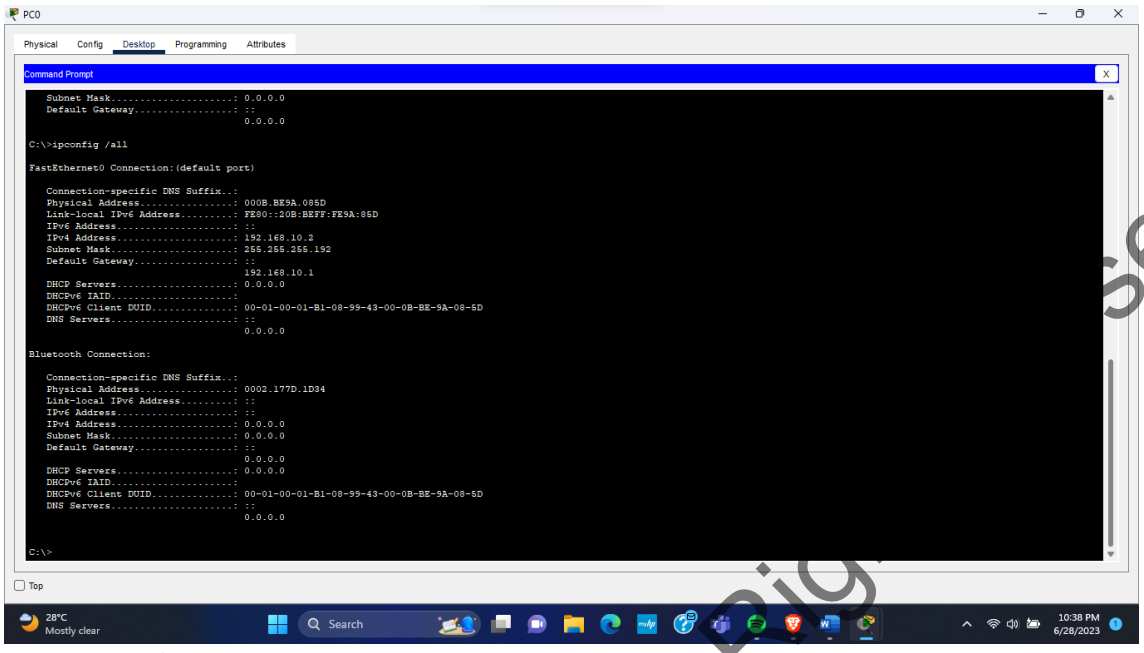


Figure 5- 9: Developer PC0

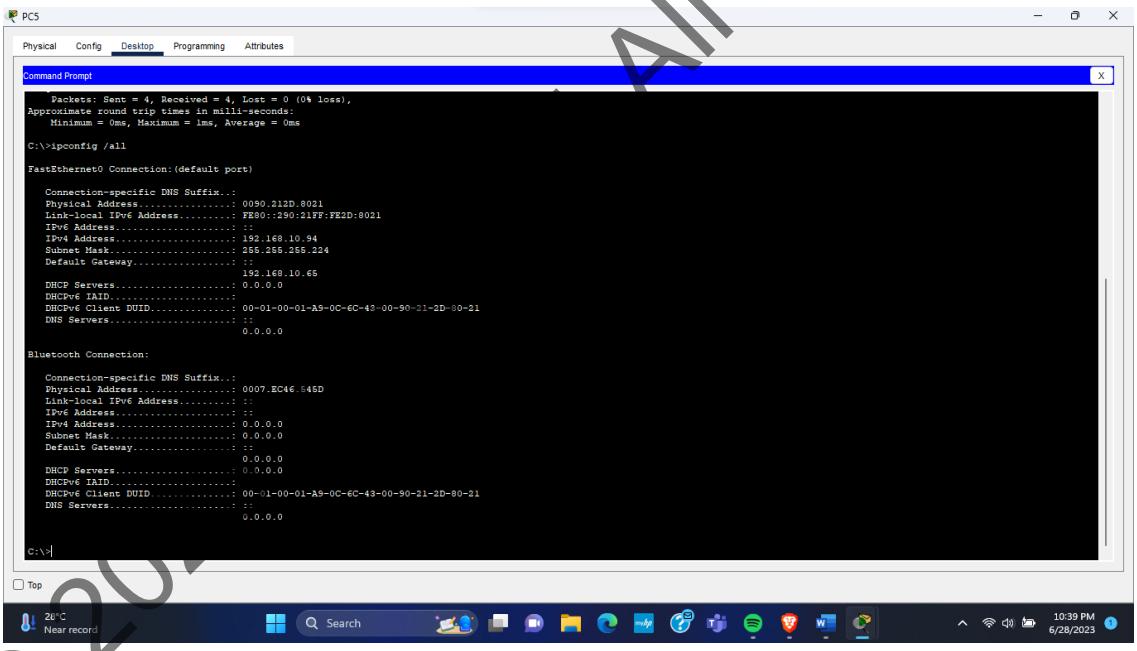


Figure 5- 10 :Finance PC5

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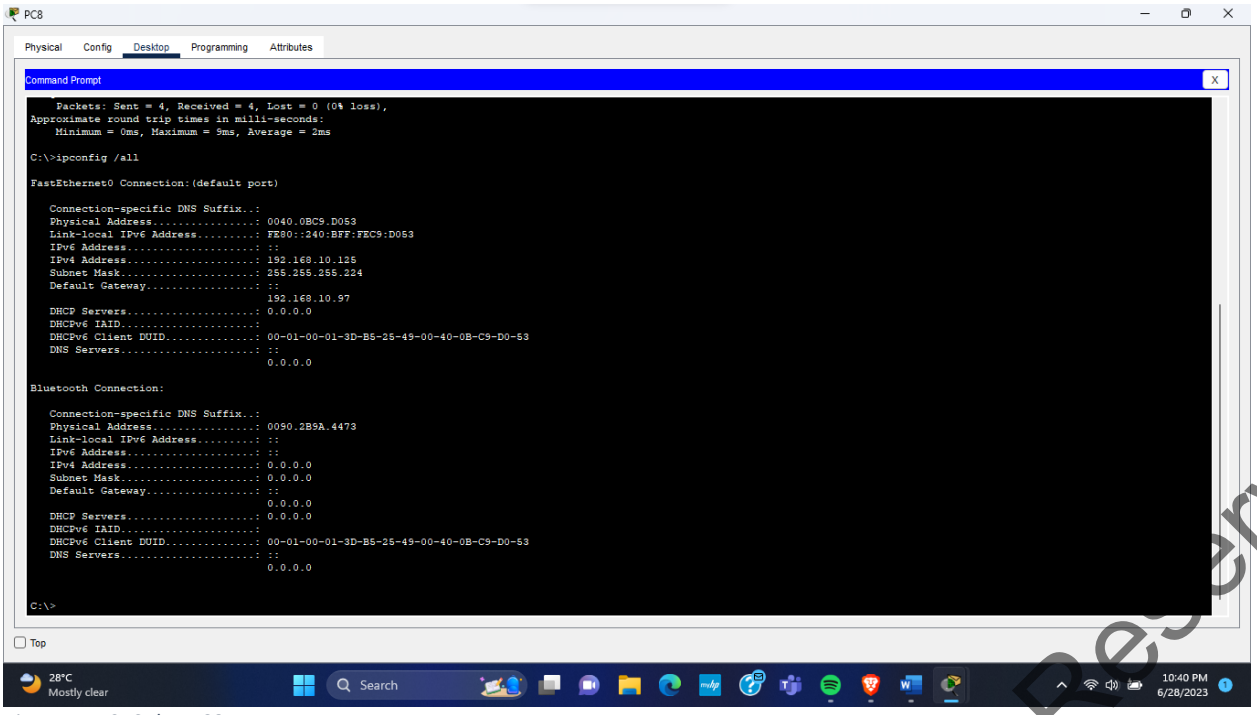


Figure 5- 10: Sales PC8

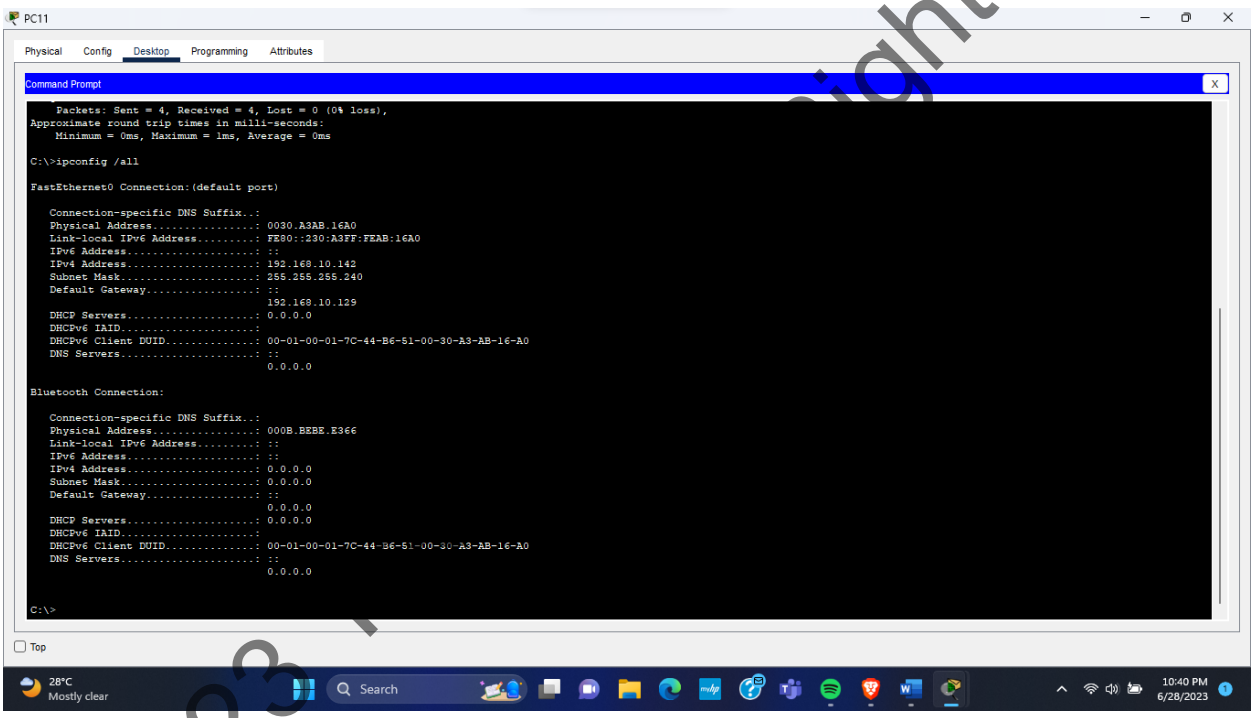


Figure 5- 11: Customer care PC11

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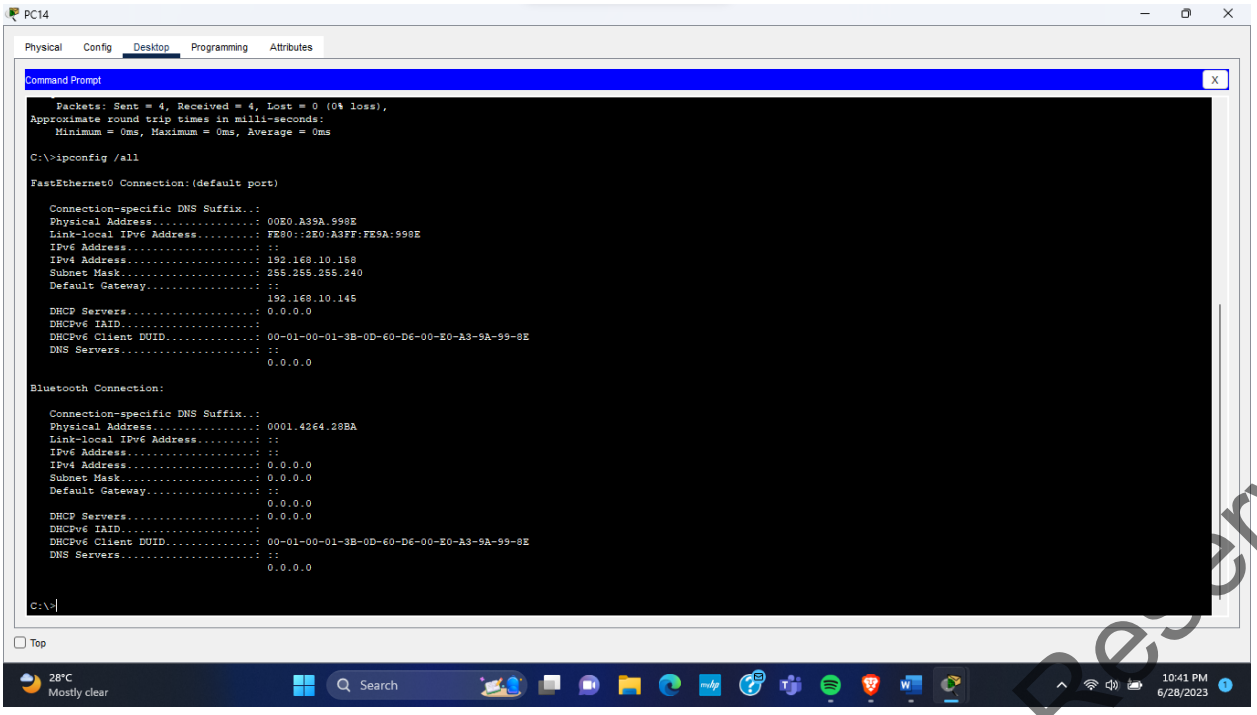


Figure 5- 12: HR PC14

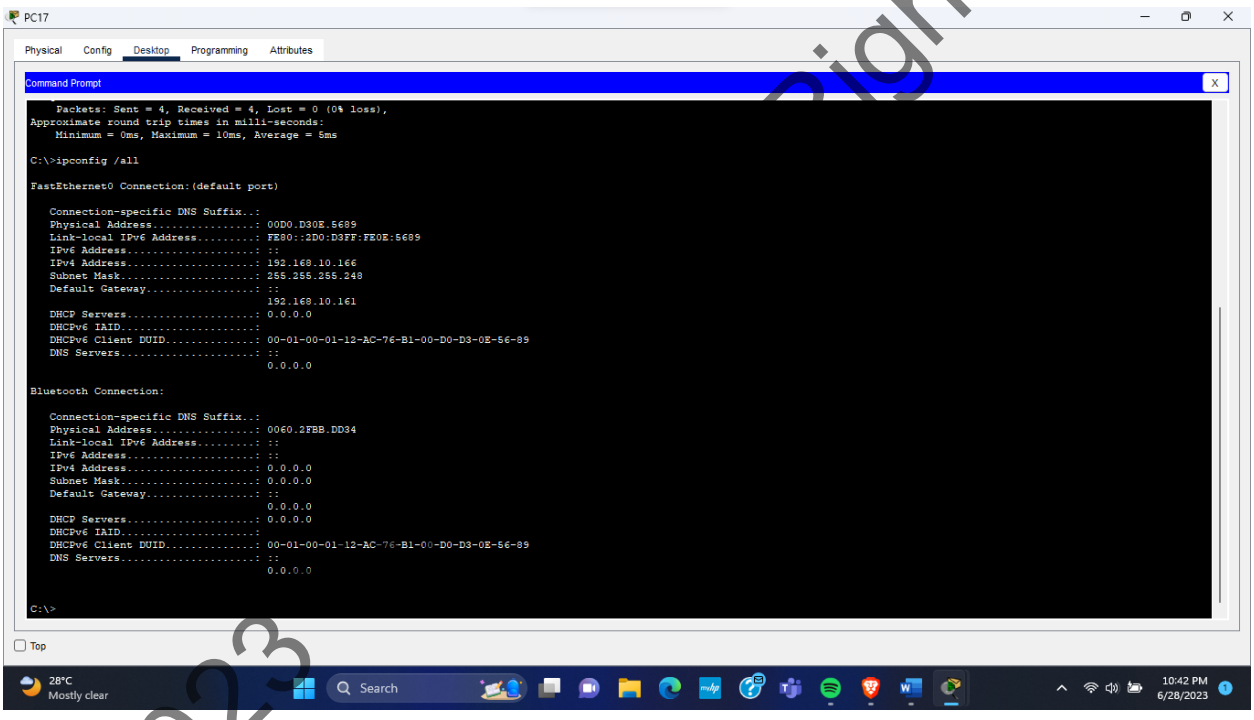


Figure 5- 13: Net Team PC17

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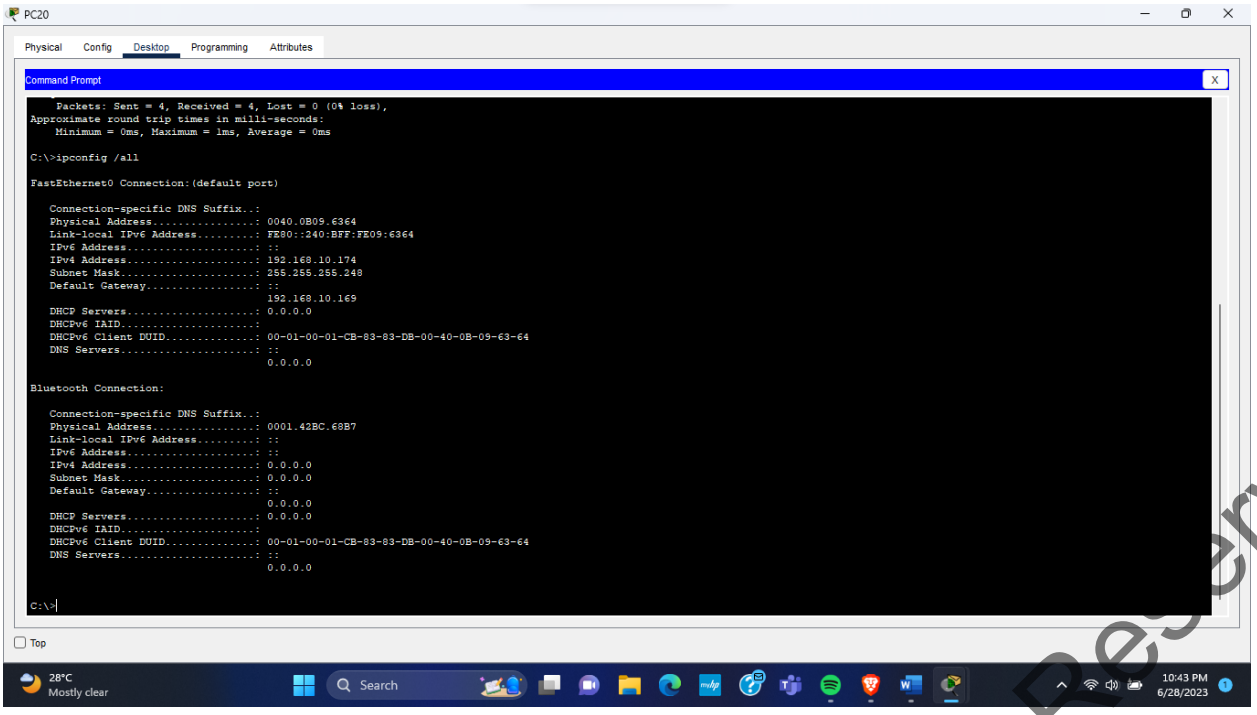


Figure 5-14: Legal PC20

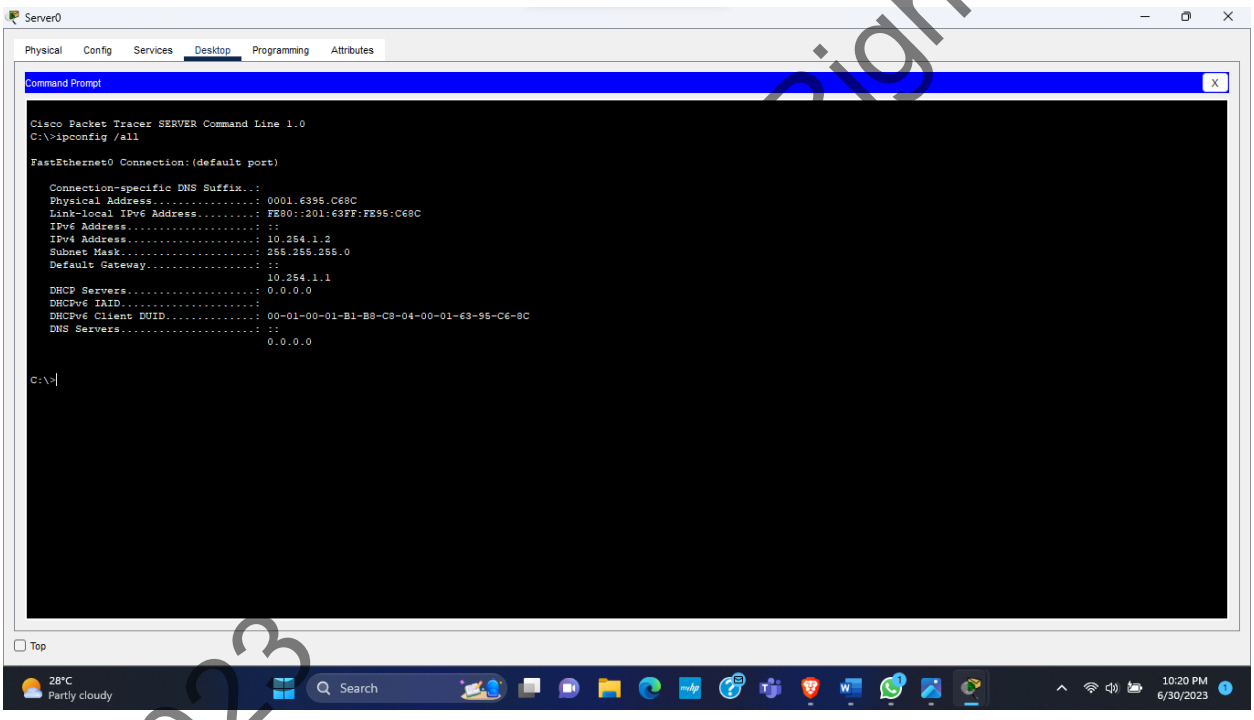


Figure 5- 15: Server Room Server0

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5.3. Traceroute Test

Used to verify the path and hops between devices in different VLANs.

5.4. Telnet Test

Perform telnet tests to check connectivity to specific services or ports between devices in different VLANs.

In below figures I have tested both testing methods at the same time

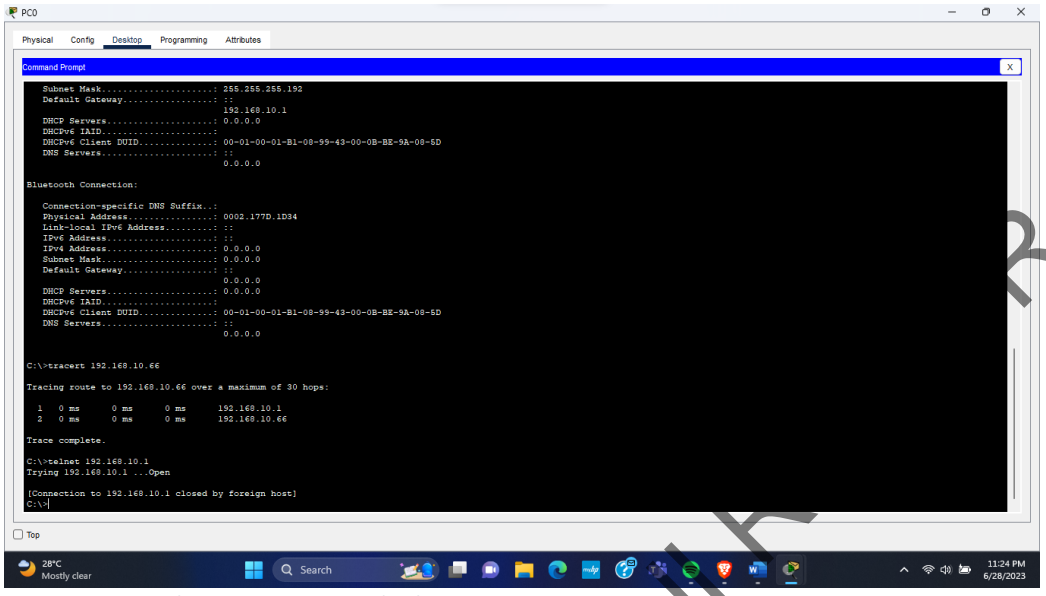


Figure 5- 16:Developer PC0 Tracert and Telnet

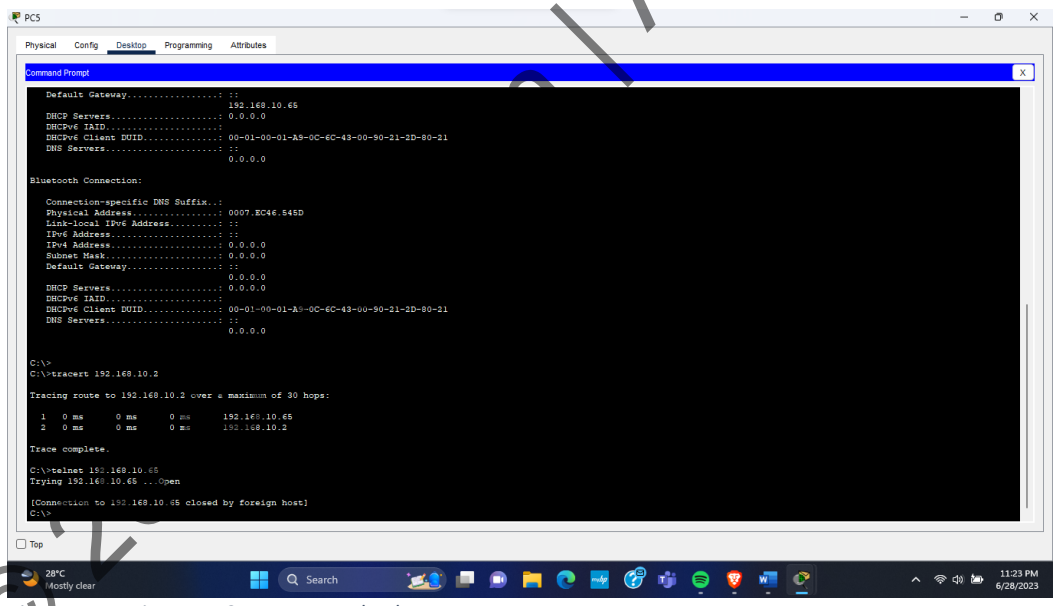


Figure 5- 17:Finance PC5 Tracert and Telnet

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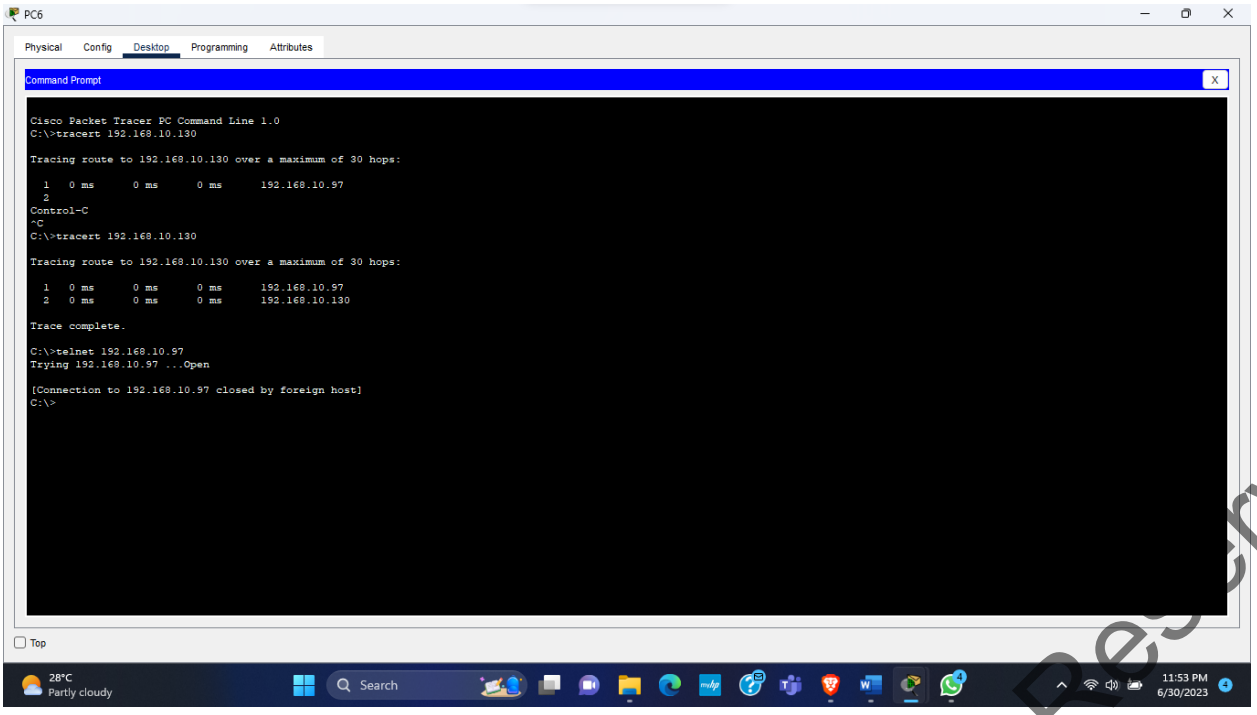


Figure 5- 18:Sales PC6 Tracert and Telnet

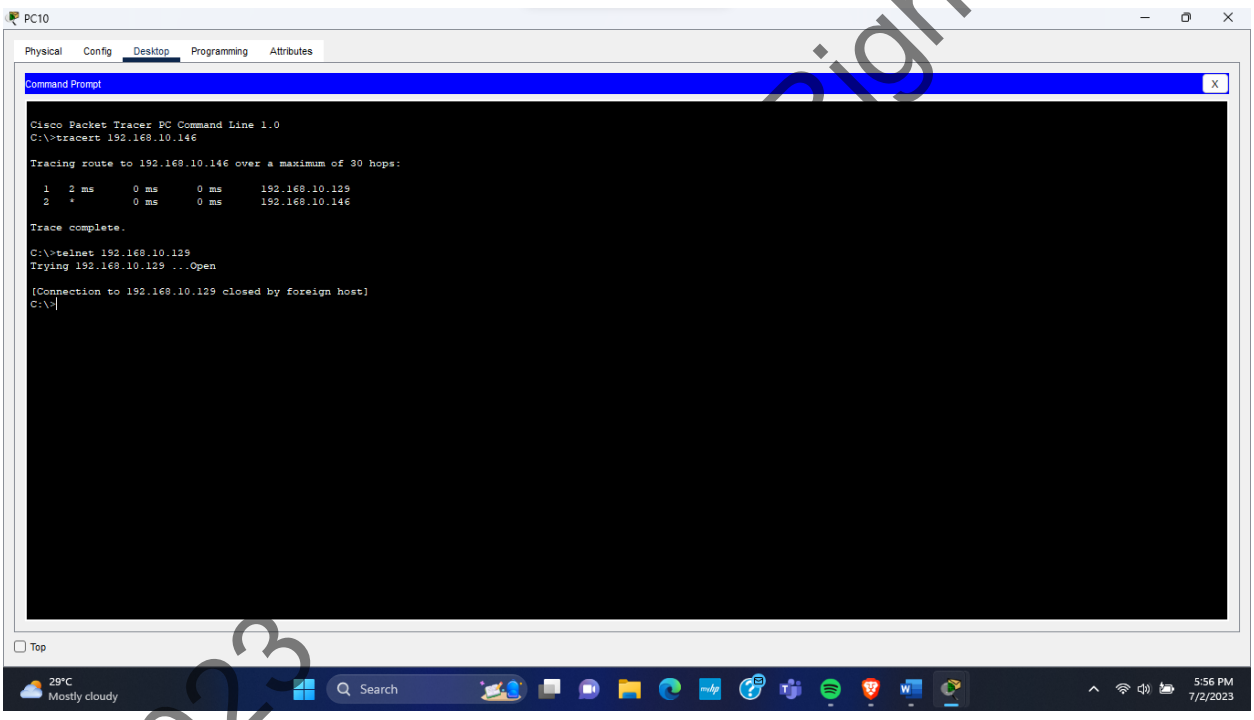


Figure 5- 19:Customer Care PC10 Tracert and Telnet

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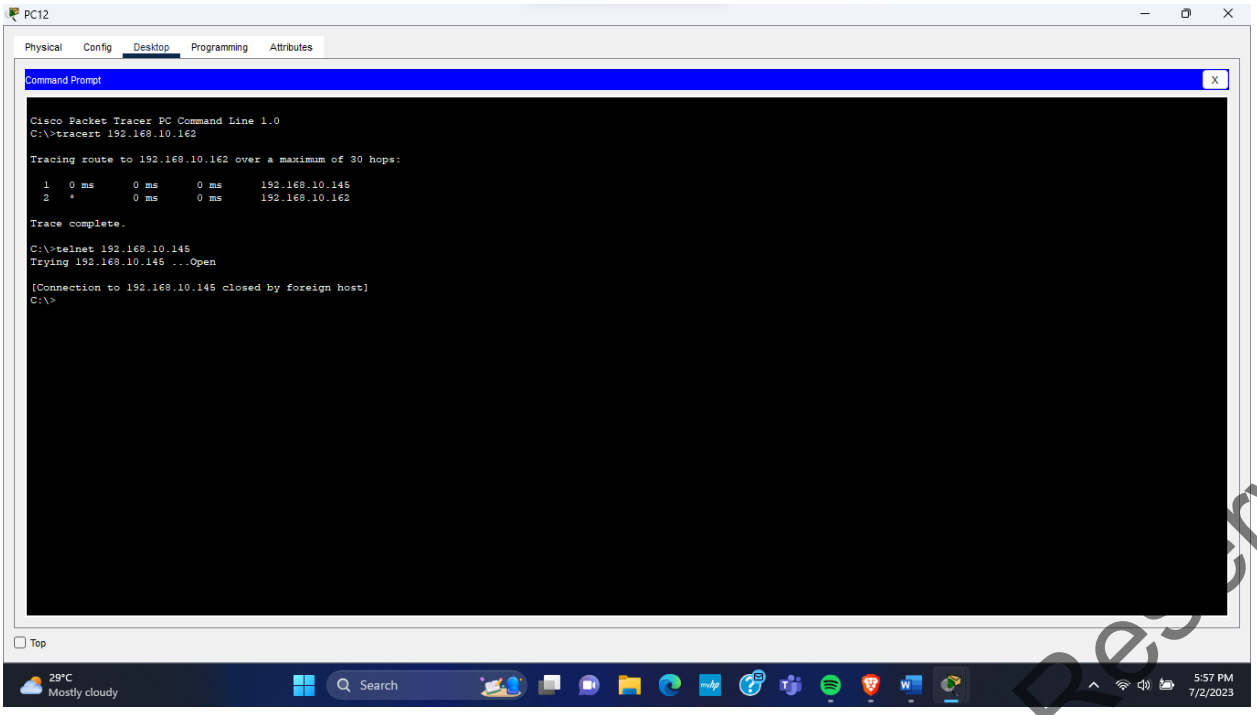


Figure 5- 20:HR PC12 Tracert and Telnet

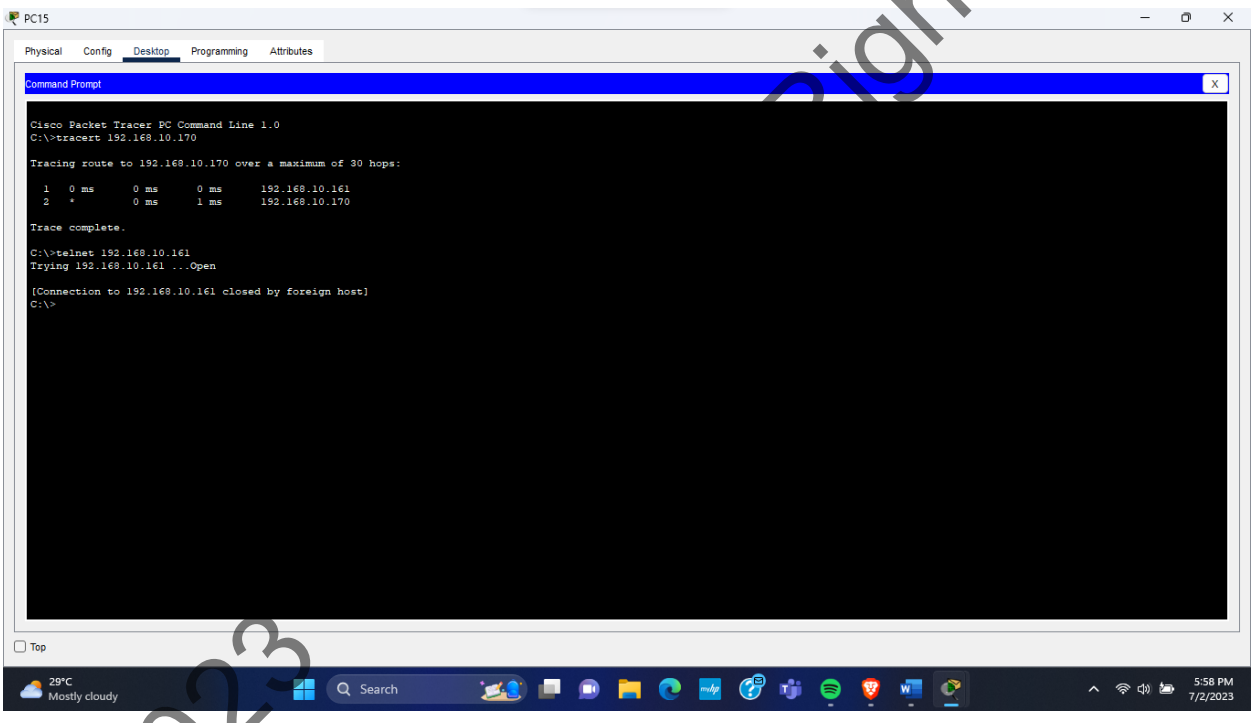


Figure 5- 21:Network Team PC15 Tracert and Telnet

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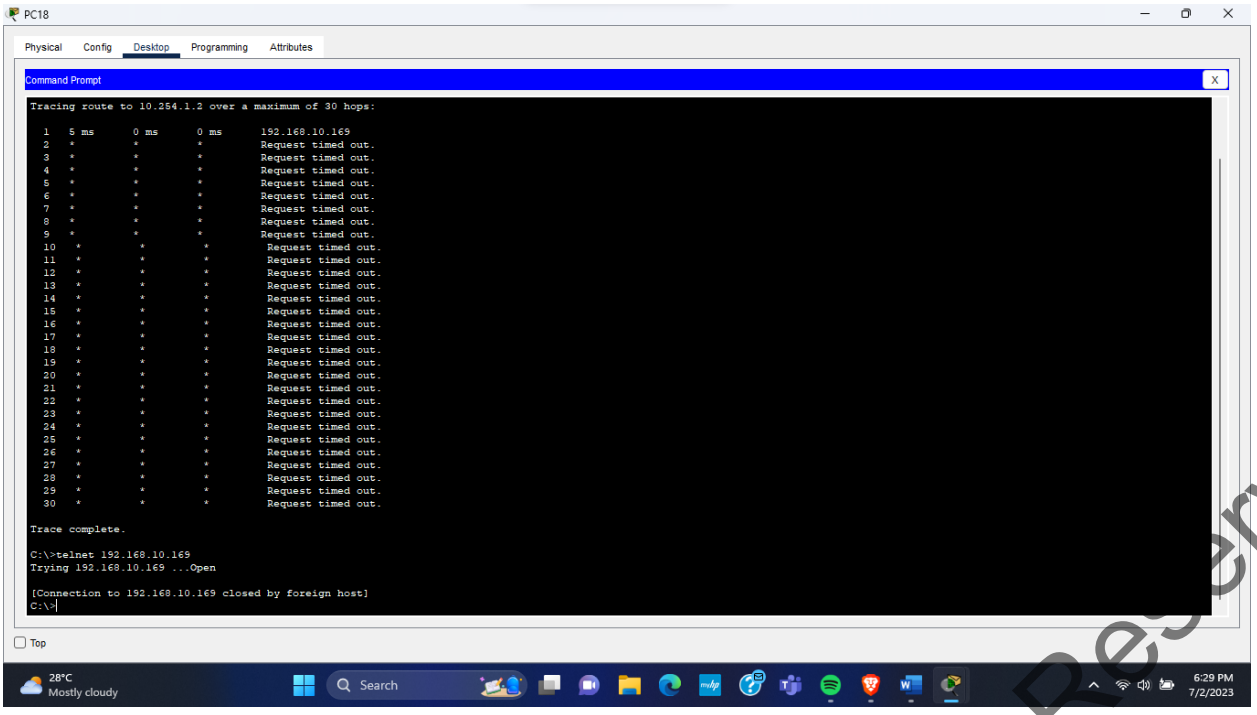


Figure 5- 22:Legal PC18 Tracert and Telnet

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6. Potential Future Enhancements

For the above implementation, below potential future enhancements can be recommended

1. **VPN Connectivity:** Implementing Virtual Private Network (VPN) connectivity would allow remote employees or branch offices to securely access the company's network resources.
2. **Wireless Network:** Setting up a wireless network infrastructure would provide flexibility and mobility for employees to connect their devices without relying on wired connections.
3. **Use of Multilayer switch (MLS)**
4. **Use of Fiber optics cable**
5. **Use of firewall**
6. **Security Enhancements:** Strengthening network security by implementing firewalls, intrusion detection and prevention systems (IDPS), and antivirus/antimalware solutions to protect against cyber threats.
7. **Cloud Integration:** Adopting cloud services for data storage, software deployment, and collaboration tools can improve scalability, data accessibility, and collaboration among employees.
8. **Disaster Recovery:** Implementing a robust backup and disaster recovery plan to ensure business continuity in case of unforeseen events, such as natural disasters or system failures.
9. **Redundancy and Resilience:** Implement redundant network connections, such as using link aggregation (e.g., LACP) or Spanning Tree Protocol (STP) to avoid single points of failure. implementing backup power solutions (e.g., uninterruptible power supply) will be an advantage to ensure network availability during power outages.

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7. Conclusion

Based on the given requirements in the scenario, following suggestions and solutions can be recommended

1. **Network Equipment:** Install managed switches that support VLANs and inter-VLAN routing capabilities ensuring the switches have enough ports to accommodate the number of users in each department.
2. **Security:** Use firewall rules or access control lists (ACLs) to allow or deny traffic based on the specific requirements and regularly update and patch network devices to protect against security vulnerabilities.
3. **Network Address Translation (NAT):** Implement NAT to provide internet access to each department while maintaining separation and configure NAT rules to translate internal departmental IP addresses to a single external IP address when accessing the internet.
4. **Network Services:** Set up a DHCP (Dynamic Host Configuration Protocol) server to assign IP addresses to devices in each department's subnet and configure DNS (Domain Name System) servers to provide name resolution for internal resources.
5. **Monitoring and Management:** Deploy network monitoring tools to monitor the health and performance of the network infrastructure. Implement centralized management solutions to simplify network administration and configuration changes.

Assumptions made for the above:

- The network switches support VLANs and inter-VLAN routing.
- The server room will host the company's servers and networking equipment.
- The DHCP and DNS servers will be set up on the network to provide IP addressing and name resolution services.
- The company has a budget for acquiring the necessary network equipment and software licenses.

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